

Quadratic Equations

Question1

If $x^2 - 4ax + 5 + a > 0$ for all $x \in R$ whenever $a \in (\alpha, \beta)$, then $4\beta + \alpha =$

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Options:

A.

0

B.

4

C.

5

D.

8

Answer: B

Solution:

$$x^2 - 4ax + 5 + a > 0$$

so, $D < 0$

$$\Rightarrow (-4a)^2 - 4(1)(5 + a) < 0$$

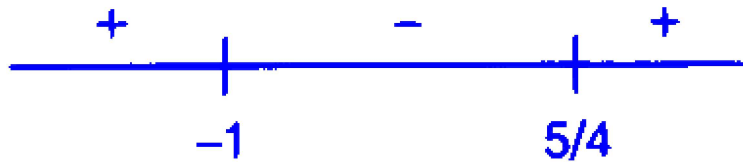
$$\Rightarrow 16a^2 - 20 - 4a < 0$$

$$\Rightarrow 4a^2 - a - 5 < 0$$

$$\Rightarrow 4a^2 - 5a + 4a - 5 < 0$$

$$\Rightarrow a(4a - 5) + 1(4a - 5) < 0$$

$$\Rightarrow (4a - 5)(a + 1) < 0$$



So, $-1 < a < \frac{3}{4}$ that means, $\alpha = -1, \beta = 5/4$

$$4\beta + \alpha = 4 \times \frac{5}{4} - 1 = 4$$

Question2

If α, β, γ are the roots of the equation $x^3 - 12x^2 + kx - 18 = 0$ and one of them is thrice the sum of the other two roots, then $\alpha^2 + \beta^2 + \gamma^2 - k =$

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Options:

A.

115

B.

41

C.

56

D.

57

Answer: D

Solution:

$$\because x^3 - 12x^2 + kx - 18 = 0$$

$$\text{So, } \alpha + \beta + \gamma = -(-12)/1 = 12 \quad \dots (i)$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = k \quad \dots (ii)$$

$$\text{and } \alpha\beta\gamma = -(-18)/1 = 18 \quad \dots (iii)$$

$$\text{Let, } \alpha = 3(\beta + \gamma) \quad \dots (iv)$$

Using Eq. (i), we get

$$\begin{aligned} 3(\beta + \gamma) + (\beta + \gamma) &= 12 \\ \Rightarrow \beta + \gamma &= 3 \end{aligned}$$

$$\text{So, } \alpha = 3 \times 3 = 9 \quad \dots (v)$$

$$\text{and } 9\beta\gamma = 18$$

$$\Rightarrow \beta\gamma = 2 \quad \dots (vi)$$

From Eqs. (v) and (vi), we get

$$\beta = 1, \gamma = 2$$

$$\begin{aligned} \text{Therefore, } k &= 9 \times 1 + 1 \times 2 + 2 \times 9 \\ &= 9 + 2 + 18 = 29 \end{aligned}$$

$$\text{Hence, } \alpha^2 + \beta^2 + \gamma^2 - k$$

$$\begin{aligned} &= 9^2 + 1^2 + 2^2 - 29 \\ &= 81 + 1 + 4 - 29 = 57 \end{aligned}$$

Question3

The polynomial equation of degree 5 whose roots are the roots of the equation $x^5 - 3x^4 - x^3 + 11x^2 - 12x + 4 = 0$ each increased by 2, is

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Options:

A.

$$x^5 - 13x^4 + 63x^3 - 135x^2 - 108x = 0$$

B.

$$x^5 - 13x^4 + 63x^3 + 135x^2 + 108x = 0$$

C.

$$x^5 - 13x^4 + 63x^3 - 135x^2 + 108x = 0$$

D.

$$x^5 - 13x^4 - 63x^3 - 135x^2 - 108x = 0$$

Answer: C

Solution:

Let

$$P(x) = x^5 - 3x^4 - x^3 + 11x^2 - 12x + 4$$

When each roots of the equation increased by 2 , then

$$P(x) \rightarrow P(x - 2)$$

$$\text{So, } P(x - 2) = (x - 2)^5 - 3(x - 2)^4$$

$$\begin{aligned} & -(x - 2)^3 + 11(x - 2)^2 - 12(x - 2) + 4 \\ &= x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32 \\ & \quad - 3(x^4 - 8x^3 + 24x^2 - 32x + 16) \\ &= (x^5 - 6x^4 + 12x^3 - 8x^2 + 11x^2 - 4x + 4) \\ & \quad - 12(x - 2) + 4 \\ &= x^5 - 13x^4 + 63x^3 - 135x^2 + 108x \end{aligned}$$

Hence, the required equation

$$x^5 - 13x^4 + 63x^3 - 135x^2 + 108x = 0$$

Question4

If the area of a square is 575 square units, then the approximate value of its side is

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Options:

A.

23.9792

B.

23.7992

C.

23.8687

D.

23.7868

Answer: A

Solution:

Let the side length as ' a '

$$\text{So, } a^2 = 575$$

$$\Rightarrow a = \sqrt{575}$$

$$\text{and } 529 < 575 < 576$$

$$\begin{aligned}\text{So, } \sqrt{575} &= 23 + \frac{575-529}{576-529} = 23 + \frac{46}{47} \\ &= 23.9792\end{aligned}$$

Question5

If the difference of the roots of the equation $x^2 - 7x + 10 = 0$ is same as the difference of the roots of the equation $x^2 - 17x + k = 0$, then a divisor of k is $x^2 - 7x + 10 = 0$

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Options:

A.

14

B.

17

C.

6

D.

15

Answer: A

Solution:

$$x^2 - 7x + 10 = 0$$

$$\therefore a = 1$$

$$\begin{aligned}\therefore \text{Difference of roots} &= \frac{\sqrt{D}}{|a|} \\ &= \frac{\sqrt{49 - 40}}{1} = \sqrt{9} = 3\end{aligned}$$

For quadratic equation

$$x^2 - 17x + k = 0$$

$$\text{Difference of roots} = \sqrt{289 - 4k}$$

According to the questions,

$$3 = \sqrt{289 - 4k}$$

Squaring both sides, we get

$$\Rightarrow 9 = 289 - 4k \Rightarrow 4k = 280$$

$$\Rightarrow k = 70$$

$$\therefore \text{Divisor of } k = 14$$

Question6

The product of all the real roots of the equation $|x|^2 - 5|x| + 6 = 0$

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Options:

A.

25

B.

36

C.

4

D.

16

Answer: B

Solution:

$$\begin{aligned}\therefore |x|^2 - 5|x| + 6 &= 0 \\ \Rightarrow |x|^2 - 2|x| - 3|x| + 6 &= 0 \\ \Rightarrow |x|(|x| - 2) - 3(|x| - 2) &= 0 \\ \Rightarrow (|x| - 2)(|x| - 3) &= 0 \\ \Rightarrow |x| = 2, |x| = 3 \\ \Rightarrow x = \pm 2, x = \pm 3 \\ \therefore \text{Product of all roots} &= 2 \times (-2) \times 3 \times (-3) \\ &= 4 \times 9 = 36\end{aligned}$$

Question7

If α, β and γ are the roots of the equation $5x^3 - 4x^2 + 3x - 2 = 0$, then $\alpha^3 + \beta^3 + \gamma^3 =$

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Options:

A.

$$\frac{17}{25}$$

B.

$$\frac{394}{125}$$

C.

$$\frac{34}{125}$$

D.

$$\frac{34}{25}$$

Answer: C

Solution:

Let the given equation be $P(x) = 5x^3 - 4x^2 + 3x - 2 = 0$.

The roots of the equation are α, β, γ .

According to Vieta's formulas, we have:

1. Sum of the roots: $\alpha + \beta + \gamma = -\left(\frac{-4}{5}\right) = \frac{4}{5}$.

2. Sum of the products of the roots taken two at a time: $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{3}{5}$.

3. Product of the roots: $\alpha\beta\gamma = -\left(\frac{-2}{5}\right) = \frac{2}{5}$.

We want to find the value of $\alpha^3 + \beta^3 + \gamma^3$.

Method 1: Using the identity for sum of cubes.

We know the identity:

$$\alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma = (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - (\alpha\beta + \beta\gamma + \gamma\alpha)).$$

First, let's find $\alpha^2 + \beta^2 + \gamma^2$:

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha).$$

Substitute the values from Vieta's formulas:

$$\alpha^2 + \beta^2 + \gamma^2 = \left(\frac{4}{5}\right)^2 - 2\left(\frac{3}{5}\right)$$

$$= \frac{16}{25} - \frac{6}{5}$$

$$= \frac{16}{25} - \frac{30}{25}$$

$$= -\frac{14}{25}.$$

Now, substitute all the required values into the sum of cubes identity:

$$\alpha^3 + \beta^3 + \gamma^3 = (\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - (\alpha\beta + \beta\gamma + \gamma\alpha)) + 3\alpha\beta\gamma.$$

$$\alpha^3 + \beta^3 + \gamma^3 = \left(\frac{4}{5}\right)\left(-\frac{14}{25} - \frac{3}{5}\right) + 3\left(\frac{2}{5}\right).$$

$$\alpha^3 + \beta^3 + \gamma^3 = \left(\frac{4}{5}\right)\left(-\frac{14}{25} - \frac{15}{25}\right) + \frac{6}{5}.$$

$$\alpha^3 + \beta^3 + \gamma^3 = \left(\frac{4}{5}\right)\left(-\frac{29}{25}\right) + \frac{6}{5}.$$

$$\alpha^3 + \beta^3 + \gamma^3 = -\frac{116}{125} + \frac{6}{5}.$$

To sum these fractions, find a common denominator, which is 125:

$$\alpha^3 + \beta^3 + \gamma^3 = -\frac{116}{125} + \frac{6 \times 25}{5 \times 25}.$$

$$\alpha^3 + \beta^3 + \gamma^3 = -\frac{116}{125} + \frac{150}{125}.$$

$$\alpha^3 + \beta^3 + \gamma^3 = \frac{150-116}{125}.$$

$$\alpha^3 + \beta^3 + \gamma^3 = \frac{34}{125}.$$

Method 2: Using Newton's Sums.

Since α, β, γ are roots of $5x^3 - 4x^2 + 3x - 2 = 0$, they satisfy the equation:

$$5\alpha^3 - 4\alpha^2 + 3\alpha - 2 = 0$$

$$5\beta^3 - 4\beta^2 + 3\beta - 2 = 0$$

$$5\gamma^3 - 4\gamma^2 + 3\gamma - 2 = 0$$

Summing these three equations:

$$5(\alpha^3 + \beta^3 + \gamma^3) - 4(\alpha^2 + \beta^2 + \gamma^2) + 3(\alpha + \beta + \gamma) - 2(1 + 1 + 1) = 0.$$

$$5(\alpha^3 + \beta^3 + \gamma^3) - 4(\alpha^2 + \beta^2 + \gamma^2) + 3(\alpha + \beta + \gamma) - 6 = 0.$$

$$\text{Let } p_1 = \alpha + \beta + \gamma = \frac{4}{5}.$$

$$\text{Let } p_2 = \alpha^2 + \beta^2 + \gamma^2. \text{ We calculated this above: } p_2 = -\frac{14}{25}.$$

$$\text{Let } p_3 = \alpha^3 + \beta^3 + \gamma^3.$$

Substitute p_1 and p_2 into the sum equation:

$$5p_3 - 4\left(-\frac{14}{25}\right) + 3\left(\frac{4}{5}\right) - 6 = 0.$$

$$5p_3 + \frac{56}{25} + \frac{12}{5} - 6 = 0.$$

To combine the constant terms, find a common denominator (25):

$$5p_3 + \frac{56}{25} + \frac{12 \times 5}{5 \times 5} - \frac{6 \times 25}{25} = 0.$$

$$5p_3 + \frac{56}{25} + \frac{60}{25} - \frac{150}{25} = 0.$$

$$5p_3 + \frac{56+60-150}{25} = 0.$$

$$5p_3 + \frac{116-150}{25} = 0.$$

$$5p_3 - \frac{34}{25} = 0.$$

$$5p_3 = \frac{34}{25}.$$

$$p_3 = \frac{34}{25 \times 5}.$$

$$p_3 = \frac{34}{125}.$$

Both methods yield the same result.

The final answer is $\boxed{\frac{34}{125}}$.

Question8

After the roots of the equation $6x^3 + 7x^2 - 4x - 2 = 0$ are diminished by h , if the transformed equation does not contain x term, then the product of all the possible value of h is

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Options:

A.

$$1/3$$

B.

$$-2/3$$

C.

$$-2/9$$

D.

$$7/3$$

Answer: C

Solution:

$$\text{Given, } 6x^3 + 7x^2 - 4x - 2 = 0$$

Let x be the root and new root $x' = x - n$

$$x = x' + n$$

$$6(x' + n)^3 + 7(x' + n)^2 - 4(x' + n) - 2 = 0$$

Coefficient of x' :

$$18n^2 + 14n - 4 = 0$$

$$\Rightarrow 9n^2 + 7n - 2 = 0$$

$$\Rightarrow 9n^2 + 9n - 2n - 2 = 0$$

$$\Rightarrow 9n(n + 1) - 2(n + 1) = 0 \Rightarrow n = -1, \frac{2}{9}$$

Product of all possible values of $n = -2/9$

Question9

The number of distinct quadratic equations $ax^2 + bx + c = 0$ with unequal real roots that can be formed by choosing the coefficients $a, b, c (a \neq b \neq c)$ from the set $\{0, 1, 2, 4\}$ is

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Options:

A.

4

B.

6

C.

5

D.

12

Answer: B

Solution:

$$ax^2 + bx + c = 0, a \neq b \neq c$$
$$a, b, c \in \{0, 1, 2, 4\}, a \neq 0$$

For unequal real roots

$$D > 0 \Rightarrow b^2 > 4ac$$

$$\text{Let } b = 4, 4ac < 16$$

$$\Rightarrow ac < 4$$

$$c = 0, a = 1, 2 \rightarrow 2$$

$$b = 2, 4ac < 4 \Rightarrow ac < 1$$

$$c = 0, a = 1, 4 \rightarrow 2$$

$$b = 1, 4ac < 1 \Rightarrow ac < \frac{1}{4}$$

$$c = 0, a = 2, 4 \rightarrow 2$$

$$\therefore \text{Total cases} = 2 + 2 + 2 = 6$$

Question10

The number of solutions of the equation $\sqrt{3x^2 + x + 5} = x - 3$ is

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Options:

A.

2

B.

1

C.

0

D.

4

Answer: C

Solution:

Given, equation is

$$\sqrt{3x^2 + x + 5} = x - 3$$

Here, $3x^2 + x + 5 \geq 0$ (since square root is non-negative)

$$\Rightarrow x - 3 \geq 0$$

$$\Rightarrow x \geq 3$$

Now, squaring to both sides of given equation, we get

$$\left(\sqrt{3x^2 + x + 5}\right)^2 = (x - 3)^2$$

$$\Rightarrow 3x^2 + x + 5 = x^2 - 6x + 9$$

$$\Rightarrow 2x^2 + 7x - 4 = 0$$

Using quadratic formula,

$$x = \frac{-7 \pm \sqrt{7^2 - 4 \cdot 2 \cdot (-4)}}{2 \cdot 2}$$

$$= \frac{-7 \pm \sqrt{49 + 32}}{4} = \frac{-7 \pm 9}{4}$$

$$\Rightarrow x = \frac{2}{4} \text{ and } x = \frac{-16}{4}$$

$$\Rightarrow x = \frac{1}{2} \text{ and } x = -4$$

Since domain, $x \geq 3$, so $x = -4$

And $x = \frac{1}{2}$ are not possible.

$\therefore$ There is zero solution.

Question11

The set of all real values of x for which $\frac{x^2-1}{(x-4)(x-3)} \geq 1$ is

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Options:

A.

$$[-1, 1] \cup (3, 4)$$

B.

$$\left[\frac{13}{7}, 3\right) \cup (4, \infty)$$

C.

$$\left(-\infty, \frac{13}{7}\right] \cup (3, 4)$$

D.

$$\mathbb{R} - [3, 4]$$

Answer: B

Solution:

$$\text{Given, } \frac{x^2-1}{(x-4)(x-3)} \geq 1$$

$$\Rightarrow \frac{x^2 - 1}{(x - 4)(x - 3)} - 1 \geq 0$$

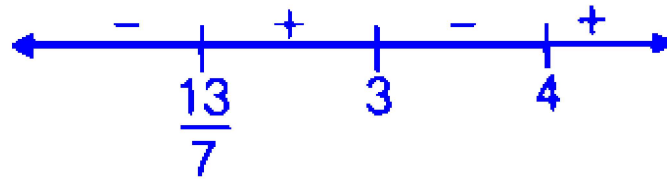
$$\Rightarrow \frac{x^2 - 1 - (x - 4)(x - 3)}{(x - 4)(x - 3)} \geq 0$$

Since, the inequality is $\frac{7x-13}{x^2-7x+12} \geq 0$

So, the critical points at $7x - 13 = 0$

$$\Rightarrow x = \frac{13}{7}$$

and at $x = 4$ and $x = 3$, the expression is undefined.



$$\therefore x \in \left[\frac{13}{7}, 3\right) \cup (4, \infty)$$

Question12

If α, β and γ are the roots of the equation $2x^3 + 3x^2 - 5x - 7 = 0$, then $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} =$

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Options:

A.

$$-\frac{17}{49}$$

B.

$$-\frac{23}{49}$$

C.

$$\frac{55}{49}$$

D.

$$\frac{67}{49}$$

Answer: D

Solution:

Given, equation is

$$2x^3 + 3x^2 - 5x - 7 = 0$$

Since, α, β and γ are the roots of the given equation.

$$\therefore \text{Sum of roots} = \alpha + \beta + \gamma = \frac{-3}{2}$$

Sum of products of roots

$$= \alpha\beta + \beta\gamma + \gamma\alpha = -\frac{5}{2}$$

$$\text{Products of roots} = \alpha\beta\gamma = -\left(\frac{-7}{2}\right) = \frac{7}{2}$$

$$\begin{aligned}\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} &= \frac{\beta^2\gamma^2 + \gamma^2\alpha^2 + \alpha^2\beta^2}{\alpha^2\beta^2\gamma^2} \\ &= \frac{(\beta\gamma^2) + (\gamma\alpha)^2 + (\alpha\beta)^2}{(\alpha\beta\gamma)^2}\end{aligned}$$

$$= \frac{(\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)}{(\alpha\beta\gamma)^2}$$

$$= \frac{\left(\frac{-5}{2}\right)^2 - 2\left(\frac{7}{2}\right)\left(\frac{-3}{2}\right)}{\left(\frac{7}{2}\right)^2}$$

$$= \frac{\frac{25}{4} + \frac{21}{2}}{\frac{49}{4}} = \frac{\frac{67}{4}}{\frac{49}{4}}$$

$$= \frac{67}{4} \times \frac{4}{49} = \frac{67}{49}$$

$$\therefore \frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} = \frac{67}{49}$$

Question13

Two roots of the equation, $ax^4 + bx^3 + cx^2 + dx + e = 0$ are positive and equal. If the product of the other two real roots is 1, then

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Options:

A.

$$be^2 = a^2d$$

B.

$$3e + \frac{2b\sqrt{e}}{\sqrt{a}} + c = a$$

C.

$$e + 2b\sqrt{e} + 3c = a\sqrt{a}$$

D.

$$b^2e = ad^2$$

Answer: B

Solution:

Given equation

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

Let $\alpha, \alpha, \beta, \frac{1}{\beta}$ be root of equation

$$\therefore \alpha + \alpha + \beta + \frac{1}{\beta} = -b/a$$

$$\Rightarrow \left(\beta + \frac{1}{\beta} \right) = -\frac{b}{a} - 2\alpha$$

$$\Rightarrow \alpha \cdot \alpha \cdot \beta \cdot \frac{1}{\beta} = \frac{e}{a} \Rightarrow \alpha^2 = e/a$$

$$\Rightarrow \alpha^2 + 2(\alpha\beta) + \frac{2\alpha}{\beta} + 1 = c/a$$

$$\alpha^2 + 2\alpha \left(\beta + \frac{1}{\beta} \right) + 1 = c/a$$

$$\Rightarrow \alpha^2 + 2\alpha \left(-\frac{b}{a} - 2\alpha \right) + 1 = c/a$$

$$\Rightarrow -3\alpha^2 - \frac{2\alpha b}{a} + 1 = c/a$$

$$\Rightarrow \frac{c}{a} + 3\alpha^2 + \frac{2\alpha b}{a} = 1$$

$$\Rightarrow 3 \left(\frac{e}{a} \right) + 2\sqrt{\frac{e}{a}} \times \frac{b}{c} + \frac{c}{a} = 1$$

$$\Rightarrow 3e + 2b\sqrt{\frac{e}{a}} + c = a$$

Question14

Let $(a - 3)x^2 + 12x + (a + 6) > 0, \forall x \in R$ and $a \in (\ell, \infty)$. If a is the least positive integral value of a , then the roots of $(\alpha - 3)x^2 + 12x + (\ell + 2) = 0$ are

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Options:

A.

1,2

B.

2,3

C.

-1, -2

D.

-2, -3

Answer: C

Solution:

Let $f(x) = (a - 3)x^2 + 12x + (a + 6)$ to quadratic always be positive

$$a - 3 > 0$$

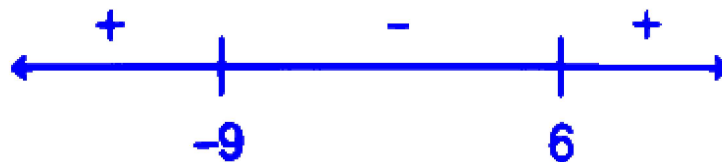
$$\Rightarrow a > 3$$

$$\text{and } D = (12)^2 - 4(a - 3)(a + 6) < 0$$
$$= 144 - 4(a^2 + 3a - 18) < 0$$

$$\Rightarrow a^2 + 3a - 54 > 0$$

$$\Rightarrow a^2 + 9a - 6a - 54 > 0$$

$$\Rightarrow (a + 9)(a - 6) > 0$$



$$a \in (-\infty, -9) \cup (6, \infty)$$

Combine with $a > 3$

$$\Rightarrow \text{We take } a \in (6, \infty)$$

Least positive integral value

$$a = 7, l = 6$$

put these values

$$\begin{aligned}
 &(\alpha - 3)x^2 + 12x + (l + 2) \\
 &= 4x^2 + 12x + 8 = 0 \\
 \Rightarrow &x^2 + 3x + 2 = 0 \\
 \Rightarrow &(x + 1)(x + 2) = 0
 \end{aligned}$$

Hence, $x = -1, -2$

Question15

If the roots of the equation $x^2 + 2ax + b = 0$ are real, distinct and differ atmost by 2 m , then b lies in the interval

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Options:

A.

$$(a^2, a^2 + m^2)$$

B.

$$(a^2 + m^2, a^2)$$

C.

$$[a^2, a^2 + 2m^2]$$

D.

$$[a^2 - m^2, a^2)$$

Answer: D

Solution:

$$\because x^2 + 2ax + b = 0$$

$$\begin{aligned}
 \Rightarrow x &= \frac{-2a \pm \sqrt{(2a)^2 - 4b}}{2} \\
 &= -a \pm \sqrt{a^2 - b}
 \end{aligned}$$

let,

$$\alpha = -a + \sqrt{a^2 - b}$$

$$\beta = -a - \sqrt{a^2 - b}$$

$$\text{so, } |\alpha - \beta| = 2\sqrt{a^2 - b}$$

$$\text{given that } 2\sqrt{a^2 - b} \leq 2m$$

$$\Rightarrow a^2 - b \leq m^2$$

$$\Rightarrow b \geq a^2 - m^2 \quad \dots (i)$$

Also, for real and distinct roots

$$D = 4a^2 - 4b > 0$$

$$\Rightarrow b < a^2 \quad \dots (ii)$$

Hence, $a^2 - m^2 \leq b$

$$\Rightarrow b \in [a^2 - m^2, a^2)$$

Question16

The cubic equation whose roots are the squares of the roots of the equation $x^3 - 2x^2 + 3x - 4 = 0$ is

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Options:

A.

$$x^3 + 2x^2 + 7x - 16 = 0$$

B.

$$x^3 + 2x^2 - 7x - 16 = 0$$

C.

$$x^3 - 2x^2 - 7x + 16 = 0$$

D.

$$x^3 - 2x^2 + 7x + 16 = 0$$

Answer: B

Solution:

$$x^3 - 2x^2 + 3x - 4 = 0$$

$$\text{So, } \alpha + \beta + \gamma = 2\alpha\beta + \beta\gamma + \gamma\alpha = +3,$$

$$\alpha\beta\gamma = 4$$

$$\text{thus, } \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2$$

$$- 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$= 4 - 2(3) = -2$$

$$\text{and } \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$$

$$= (\alpha\beta)^2 + (\beta\gamma)^2 + (\gamma\alpha)^2$$

$$\therefore \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$$

$$= (\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

$$= (3)^2 - 2(4)(2) = 9 - 16 = -7$$

$$\text{and } \alpha^2\beta^2\gamma^2 = (\alpha\beta\gamma)^2 = 4^2 = 16$$

Hence, the cubic equation with roots $\alpha^2, \beta^2, \gamma^2$ is

$$x^3 - (\alpha^2 + \beta^2 + \gamma^2)x^2 + (\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2)x - \alpha^2\beta^2\gamma^2$$

$$= x^3 - (-2)x^2 + (-7)x - 16 \quad x - \alpha^2\beta^2\gamma^2$$

$$= x^3 + 2x^2 - 7x - 16 = 0$$

Question17

If α, β, γ are the roots of the equation $x^3 + px^2 + qx + r = 0$, then $\alpha^3 + \beta^3 + \gamma^3 =$

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Options:

A.

$$p^3 - 3pq + r$$

B.

$$p^2 - 2pq + r$$

C.

$$3pq - 3r - p^3$$

D.

$$3pq + 3r + p^3$$

Answer: C

Solution:

$$\because x^3 + px^2 + qx + r = 0$$

$$\alpha + \beta + \gamma = -p, \alpha\beta + \beta\gamma + \gamma\alpha = q,$$

$$\alpha\beta\gamma = -r$$

$$\begin{aligned} \text{So, } \alpha^3 + \beta^3 + \gamma^3 &= (\alpha + \beta + \gamma)^3 \\ &\quad - 3(\alpha + \beta + \gamma)(\alpha\beta + \beta\gamma + \gamma\alpha) + 3\alpha\beta\gamma \\ &= (-p)^3 - 3(-p)(q) + 3(-r) \\ &= -p^3 + 3pq - 3r \end{aligned}$$

Question18

If α, β are the roots of the equation $x^2 + bx + c = 0$ satisfying the conditions $\alpha + \beta = 5$ and $\alpha^3 + \beta^3 = 60$, then $3c + 2 =$

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Options:

A.

$$2b$$

B.

$$3b$$

C.

$$-3b$$

D.

$$-2b$$

Answer: C

Solution:

$$\therefore \alpha + \beta = 5 \Rightarrow \frac{-b}{1} = 5 \Rightarrow b = -5 \dots (i)$$

$$\text{and } \alpha^3 + \beta^3 = 60$$

$$\Rightarrow (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = 60$$

$$\Rightarrow (5)^3 - 3 \times c(5) = 60$$

$$\Rightarrow 125 - 15c = 60$$

$$\Rightarrow 15c = 125 - 60 = 65$$

$$\Rightarrow c = \frac{65}{15} = \frac{13}{3}$$

$$\therefore 3c + 2 = 3 \times \frac{13}{3} + 2 = 13 + 2 = 15$$

$$= (-3)(-5) = -3b \text{ (from Eq. (i))}$$

Question19

If α, β, γ are the roots of the equation,

$$x^3 + ax^2 + bx + c = 0, \text{ then } (\alpha + \beta - 2\gamma) \\ (\beta + \gamma - 2\alpha)(\gamma + \alpha - 2\beta) =$$

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Options:

A.

$$2a^3 + 9ab + 27c$$

B.

$$2a^3 + 9ab - 27c$$

C.

$$2a^3 - 9ab - 27c$$

D.

$$2a^3 - 9ab + 27c$$

Answer: D

Solution:

Given, α, β and γ are roots of equation

$$x^3 + ax^2 + bx + c = 0$$

$$\alpha + \beta + \gamma = -a$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = b$$

$$\alpha\beta\gamma = -c$$

$$\text{Now, } (\alpha + \beta - 2\gamma)(\beta + \gamma - 2\alpha)(\gamma + \alpha - 2\beta)$$

$$= (\alpha + \beta + \gamma - 3\gamma)(\alpha + \beta + \gamma - 3\alpha)(\alpha + \beta + \gamma - 3\beta)$$

$$= -(\alpha + 3\alpha)(\alpha + 3\beta)(\alpha + 3\gamma)$$

$$= -(a^3 + 3(\alpha + \beta + \gamma)a^2 + 9(\alpha\beta + \beta\gamma + \gamma\alpha)a + 27(\alpha\beta\gamma))$$

$$= -(a^3 - 3a^3 + 9ab - 27c)$$

$$= 2a^3 - 9ab + 27c$$

Question20

If the sum of two roots of the equation $x^4 + 2x^3 - 7x^2 - 8x + 12 = 0$ is zero, then the sum of the squares of the other two roots is

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Options:

A.

5

B.

10

C.

13

D.

25

Answer: B

Solution:

Given,

$$\begin{aligned}
 x^4 + 2x^3 - 7x^2 - 8x + 12 &= 0 \\
 \Rightarrow (x-1)(x-3)(x-2)(x+2) &= 0 \\
 \Rightarrow x = 1, 3, 2, -2
 \end{aligned}$$

Since sum of two roots of Eq. (i) is zero.

$$\text{So, } -2 + 2 = 0$$

Hence, sum of the squares of the other two roots

$$(1)^2 + (3)^2 = 10$$

Question 21

$f(x)$ is a quadratic polynomial satisfying the condition $f(x) + f\left(\frac{1}{x}\right) = f(x)f\left(\frac{1}{x}\right)$. If $f(-1) = 0$, then the range of f is

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Options:

A.

$$[1, \infty)$$

B.

$$[-1, 1]$$

C.

$$(-\infty, 1]$$

D.

$$\mathbb{R}$$

Answer: C

Solution:

$$f(x) = ax^2 + bx + c$$

substitute $f(x)$ into the given equation

$$\begin{aligned}
& ax^2 + bx + c + a\left(\frac{1}{x}\right)^2 + b\left(\frac{1}{x}\right) + c \\
&= (ax^2 + bx + c) \left(a\left(\frac{1}{x}\right)^2 + b\left(\frac{1}{x}\right) + c \right) \\
&\Rightarrow ax^2 + bx + c + \frac{a}{x^2} + \frac{b}{x} + c \\
&= (ax^2 + bx + c) \left(\frac{a}{x^2} + \frac{b}{x} + c \right) \\
&\Rightarrow ax^4 + bx^3 + cx^2 + a + bx + cx^2 \\
&= (ax^2 + bx + c) (a + bx + cx^2) \\
&\Rightarrow ax^4 + bx^3 + 2cx^2 + bx + a \\
&= a^2x^2 + abx^3 + acx^4 + abx + b^2x^2 + bcx^3 + ac + bcx + c^2x^2
\end{aligned}$$

Compare coefficients : $a = ac$,

$$b = ab + bc$$

$$2c = a^2 + b^2 + c^2, b = ab + bc, a = ac$$

From $a = ac$, we have $c = 1$ or $a = 0$, if $a = 0$ then $f(x)$ is not quadratic, so $c = 1$ from $b = ab + bc$, we have $b = ab + b$, so $ab = 0$, since $a \neq 0, b = 0$

From $2c = a^2 + b^2 + c^2$, we have

$$2 = a^2 + 0 + 1$$

$$\text{So, } a^2 = 1 \text{ and } a = \pm 1$$

$$\text{thus, } f(x) = x^2 + 1 \text{ or } f(x) = -x^2 + 1$$

$$\text{Since, } f(-1) = 0, \text{ we have } f(x) = -x^2 + 1$$

$$\Rightarrow f(x) = -x^2 + 1$$

$$\Rightarrow \text{Range} = (-\infty, 1]$$

Question22

If $\alpha \neq 0$ and zero are the roots of the equation $x^2 - 5kx + (6k^2 - 2k) = 0$, then $\alpha =$

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Options:

A.

$$\frac{1}{3}$$

B.

1

C.

$$\frac{5}{3}$$

D.

5

Answer: C

Solution:

If $\alpha \neq 0$ and 0 are the roots of

$$x^2 - 5kx + (6k^2 - 2k) = 0, \text{ then find } \alpha.$$

Step 1: Use the fact that 0 is a root.

If $x = 0$ is a root, it should satisfy the equation:

$$0^2 - 5k(0) + (6k^2 - 2k) = 0$$

$$\Rightarrow 6k^2 - 2k = 0$$

$$\Rightarrow 2k(3k - 1) = 0$$

$$\text{So } k = 0 \text{ or } k = \frac{1}{3}.$$

But if $k = 0$, the equation becomes $x^2 = 0$, giving a double root 0 — contradicting that one root is nonzero ($\alpha \neq 0$).

Hence,

$$k = \frac{1}{3}.$$

Step 2: Substitute $k = \frac{1}{3}$ into the equation.

$$x^2 - 5\left(\frac{1}{3}\right)x + \left(6\left(\frac{1}{3}\right)^2 - 2\left(\frac{1}{3}\right)\right) = 0$$

$$\Rightarrow x^2 - \frac{5}{3}x + \left(\frac{6}{9} - \frac{2}{3}\right) = 0$$

$$\Rightarrow x^2 - \frac{5}{3}x + \left(\frac{2}{3} - \frac{2}{3}\right) = 0$$

Wait — compute the constant term carefully:

$$\frac{6}{9} - \frac{2}{3} = \frac{2}{3} - \frac{2}{3} = 0.$$

So the equation becomes:

$$x^2 - \frac{5}{3}x = 0$$

$$x\left(x - \frac{5}{3}\right) = 0$$

Step 3: Identify the roots.

Roots are $x = 0$ and $x = \frac{5}{3}$.

Given that one root is 0 and another is $\alpha \neq 0$,

$$\alpha = \frac{5}{3}.$$

 **Final Answer:**

$$\alpha = \frac{5}{3}$$

Option C.

Question23

The set of all real values of x satisfying the inequation $\frac{8x^2-14x-9}{3x^2-7x-6} > 2$ is

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Options:

A.

$$(-\infty, 1) \cup (3, \infty)$$

B.

$$(-\infty, -\frac{2}{3}) \cup (2, \infty)$$

C.

$$(-\frac{2}{3}, 2)$$

D.

$$(-\infty, -\frac{2}{3}) \cup (3, \infty)$$

Answer: D

Solution:

$$\begin{aligned} \frac{8x^2 - 14x - 9}{3x^2 - 7x - 6} - 2 &> 0 \\ \Rightarrow \frac{8x^2 - 14x - 9 - 6x^2 + 14x + 12}{3x^2 - 7x - 6} &> 0 \\ \Rightarrow \frac{2x^2 + 3}{3x^2 - 9x + 2x - 6} &> 0 \\ \Rightarrow \frac{2x^2 + 3}{3x(x - 3) + 2(x - 3)} &> 0 \\ \Rightarrow \frac{2x^2 + 3}{(x - 3)(3x + 2)} &> 0 \end{aligned}$$



$$x \in \left(-\infty, -\frac{2}{3}\right) \cup (3, \infty)$$

Question24

When the roots of $x^3 + \alpha x^2 + \beta x + 6 = 0$ are increased by 1, if one of the resultant values is the least root of $x^4 - 6x^3 + 11x^2 - 6x = 0$, then

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Options:

A.

$$\alpha - \beta + 5 = 0$$

B.

$$\alpha + \beta + 7 = 0$$

C.

$$2\alpha + \beta + 7 = 0$$

D.

$$2\alpha + 3\beta - 1 = 0$$

Answer: A

Solution:

$$x(x^3 - 6x^2 + 11x - 6) = 0$$

$$\Rightarrow x(x-1)(x-2)(x-3) = 0$$

$$\Rightarrow x = 0, 1, 2, 3$$

The least roots is 0 .

Let the roots of $x^3 + \alpha x^2 + \beta x + 6 = 0$ be r_1, r_2, r_3

The roots of $x^3 + \alpha x^2 + \beta x + 6 = 0$

increased by 1 are $r_1 + 1, r_2 + 1, r_3 + 1$

One of these roots is 0 , so $r_i + 1 = 0$. for some i^0 .

$$\Rightarrow r_i = -1$$

Since $r_i = -1$ is root of

$$x^3 + \alpha x^2 + \beta x + 6 = 0$$

$$\Rightarrow (-1)^3 + \alpha(-1)^2 + \beta(-1) + 6 = 0$$

$$\Rightarrow -1 + \alpha - \beta + 6 = 0$$

$$\Rightarrow \alpha - \beta + 5 = 0$$

Question25

Let ' a ' be a non-zero real number. If the equation whose roots are the squares of the roots of the cubic equation $x^3 - ax^2 + ax - 1 = 0$ is identical with this cubic equation, then ' a ' =

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Options:

A.

$$\frac{1}{3}$$

B.

$$3$$

C.

$$\frac{1}{2}$$

D.

Answer: B

Solution:

$$\alpha + \beta + \gamma = a$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = a$$

$$\alpha\beta\gamma = 1$$

$$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$\Rightarrow \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = (\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

$$= a^2 - 2a$$

$$\therefore \alpha^2\beta^2\gamma^2 = (\alpha\beta\gamma)^2 = 1^2 = 1$$

Now, equate the coefficients of the original and new equations.

$$a^2 - 2a = a$$

$$a^2 - 2a = a$$

$$1 = 1$$

Solve for a

$$a^2 - 2a = a$$

$$\Rightarrow a^2 - 3a = 0$$

$$\Rightarrow a(a - 3) = 0$$

$$\therefore a \neq 0 \therefore a = 3$$

Question26

If $(2k - 1)x^2 - 2(3k - 2)x + 4k > 0$ for every $x \in R$, then the sum of all possible integral values of k is

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Options:

A.

21

B.

27

C.

36

D.

28

Answer: D

Solution:

We have,

$$(2k-1)x^2 - 2(3k-2)x + 4k > 0, \forall x \in \mathbb{R}$$

$$\Rightarrow 4(3k-2)^2 - 4(2k-1)4k < 0$$

$$\Rightarrow 9k^2 - 12k + 4 - 8k^2 + 4k < 0$$

$$\Rightarrow k^2 - 8k + 4 < 0,$$

$$k = \frac{8 \pm \sqrt{64-16}}{2} = \frac{8 \pm 4\sqrt{3}}{2}$$

$$\Rightarrow k \in (4 - 2\sqrt{3}, 4 + 2\sqrt{3})$$

$$k = 1, 2, 3, \dots, 7$$

$$\therefore \text{Sum of integral value} = 28$$

Question27

If α is a repeated root of multiplicity 2 of the equation $18x^3 - 33x^2 + 20x - 4 = 0$, then

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Options:

A.

$$3\alpha^2 - 8\alpha + 4 = 0$$

B.

$$3\alpha^2 + 8\alpha + 4 = 0$$

C.

$$3\alpha^2 - \alpha - 4 = 0$$

D.

$$3\alpha^2 + 2\alpha - 4 = 0$$

Answer: A

Solution:

$$f(x) = 18x^3 - 33x^2 + 20x - 4 = 0$$

α is a repeated root of multiplicity ? Let roots be α, α, β

$$\begin{aligned} f'(x) = 54x^2 - 66x + 20 = 0 \text{ is repeated roots, } f'(\alpha) = 0 & \Rightarrow 54\alpha^2 - 66\alpha + 20 = 0 \\ & \Rightarrow 27\alpha^2 - 33\alpha + 10 = 0 \\ & \Rightarrow 27\alpha^2 - 18\alpha - 15\alpha + 10 = 0 \\ & \Rightarrow (9\alpha - 5)(3\alpha - 2) = 0 \Rightarrow \alpha = \frac{2}{3} \text{ or } \frac{5}{9} \end{aligned}$$

$$\text{So, } \alpha = \frac{2}{3} \text{ and } 27\alpha^2 - 8\alpha + 4 = 0$$

Question 28

The equation $6x^4 - 5x^3 + 13x^2 - 5x + 6 = 0$ will have

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Options:

A.

only real roots

B.

only complex roots

C.

two real and two complex roots

D.

two real and two purely imaginary roots

Answer: B

Solution:

We start with the equation: $6x^4 - 5x^3 + 13x^2 - 5x + 6 = 0$.

Divide both sides by x^2 (where $x \neq 0$) to get:

$$6x^2 - 5x + 13 - \frac{5}{x} + \frac{6}{x^2} = 0$$

Rewrite the expression by grouping terms: $6\left(x^2 + \frac{1}{x^2}\right) - 5\left(x + \frac{1}{x}\right) + 13 = 0$

Let $t = x + \frac{1}{x}$. Then $x^2 + \frac{1}{x^2} = t^2 - 2$.

Substitute these into the equation: $6(t^2 - 2) - 5t + 13 = 0$

Simplify: $6t^2 - 12 - 5t + 13 = 0$ $6t^2 - 5t + 1 = 0$

This is a quadratic equation in t . The discriminant (D) is:

$$D = (-5)^2 - 4 \cdot 6 \cdot 1 = 25 - 24 = 1 \text{ (which is greater than 0).}$$

This means there are two real solutions for t : $t = \frac{5 \pm 1}{12} = \frac{6}{12}$ or $\frac{4}{12}$ $t = \frac{1}{2}$ or $\frac{1}{3}$

So, $x + \frac{1}{x} = \frac{1}{2}$ or $x + \frac{1}{x} = \frac{1}{3}$.

Now solve for x in each case:

If $x + \frac{1}{x} = \frac{1}{2}$, multiply both sides by x : $x^2 + 1 = \frac{1}{2}x$. Rearranged: $2x^2 - x + 2 = 0$

If $x + \frac{1}{x} = \frac{1}{3}$, multiply both sides by x : $x^2 + 1 = \frac{1}{3}x$. Rearranged: $3x^2 - x + 3 = 0$

For both equations, the discriminant is less than 0. This means there are no real solutions for x .

Therefore, all the solutions for x are complex numbers.

Question 29

The roots α, β of the equation $x^2 - 6(k - 1)x + 4(k - 2) = 0$ are equal in magnitude but opposite in sign, if $\alpha > \beta$, then the product of the roots of the equation $2x^2 - \alpha x + 6\beta(\alpha + 1) = 0$

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Options:

A.

12

B.

-12

C.

16

D.

-18

Answer: D

Solution:

$$\text{Given, } x^2 - 6(k-1)x + 4(k-2) = 0$$

$$\alpha + \beta = 6(k-1) = 0 \Rightarrow k = 1$$

$$\alpha\beta = 4(k-2) < 0 \Rightarrow k < 2$$

$$\text{for } k = 1, x^2 - 4 = 0 \Rightarrow x^2 = 4$$

$$\therefore x = \pm 2 \Rightarrow \alpha = +2 \text{ and } \beta = -2$$

Then, $2x^2 - \alpha x + 6\beta(\alpha + 1) = 0$ become

$$2x^2 - 2x + 6(-2)(3) = 0$$

$$x^2 - x - 18 = 0$$

Now, product of roots = -18

Question30

If $ax^2 + bx + c < 0 \forall x \in R$ and the expressions $cx^2 + ax + b$ and $ax^2 + bx + c$ have their extreme values at the same point x , then for the expression $cx^2 + ax + b$

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Options:

A.

$$\text{Minimum value} = \frac{4b}{3}$$

B.

$$\text{Maximum value} = \frac{4a}{3}$$

C.

$$\text{Minimum value} = \frac{3a}{4}$$

D.

$$\text{Maximum value} = \frac{3b}{4}$$

Answer: D

Solution:

We have, $ax^2 + bx + c < 0$

If $a < 0$, then $\Delta < 0$

$$a < 0 \text{ and } b^2 - 4ac < 0$$

$$\therefore b^2 < 4ac$$

And $cx^2 + ax + b$ and $ax^2 + bx + c$ have same extreme value

$$\therefore -\frac{a}{2c} = -\frac{b}{2a} \Rightarrow a^2 = bc$$

$$\text{Now, the minimum value of } cx^2 + ax + b \text{ is } -\frac{\Delta}{4a} = -\frac{a^2 - 4bc}{4c} = -\frac{bc - 4bc}{4c} = -\left(-\frac{3bc}{4c}\right) = \frac{3b}{4}$$

Because $b^2 = \text{positive}$ and $a < 0 \Rightarrow c < 0$

$$\therefore \frac{3b}{4} \text{ is the maximum value of } cx^2 + ax + b$$

Question31

If $x^2 - 5x + 6$ is a factor of $f(x) = x^4 - 17x^3 + kx^2 - 247x + 210$, then the other quadratic factor of $f(x)$ is

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Options:

A.

$$x^2 + 12x + 35$$

B.

$$x^2 - 12x + 35$$

C.

$$x^2 - 6x + 35$$

D.

$$x^2 + 6x + 35$$

Answer: B

Solution:

$$f(x) = x^4 - 17x^3 + kx^2 - 247x + 210$$

Its factor: $x^2 - 5x + 6$

$\therefore (x - 2)(x - 3)$ is also a factor.

$$f(2) = 16 - 17(8) + k(4) - (247)2 + 210 = 0$$

$$\Rightarrow 16 - 136 + 4k - 494 + 210 = 0$$

$$\Rightarrow 4k - 404 = 0 \Rightarrow k = 101$$

$$\therefore f(x) = x^4 - 17x^3 + 101x^2 - 247x + 210 \\ = (x^2 - 5x + 6)(x^2 - 12x + 35)$$

Question32

Given $f(x) = x^2 - 5x + 4$. Out of first 20 natural numbers, if a number x is chosen at random, then the probability that the chosen x satisfies the inequality $f(x) > 10$ is

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Options:

A.

$$\frac{1}{2}$$

B.

$$\frac{3}{4}$$

C.

$$\frac{7}{10}$$

D.

$$\frac{13}{20}$$

Answer: C

Solution:

$$\text{Given, } f(x) = x^2 - 5x + 4$$

$$f(x) > 10$$

$$x^2 - 5x + 4 > 10$$

$$x^2 - 5x - 6 > 0$$

$$(x - 6)(x + 1) > 0$$

$$x < -1 \text{ or } x > 6$$

$$n(S) = {}^{20}C_1 = 20$$

$$n(E) = {}^{14}C_1 = 14$$

$$P(E) = \frac{14}{20} = \frac{7}{10}$$

Question33

If the harmonic mean of the roots of the equation $\sqrt{2}x^2 - bx + (8 - 2\sqrt{5}) = 0$ is

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Options:

A.

$$3$$

B.

$$2$$

C.

$$4 - \sqrt{5}$$

D.

$$4 + \sqrt{5}$$

Answer: C

Solution:

$$\sqrt{2}x^2 - bx + (8 - 2\sqrt{5}) = 0$$

Let its roots be α and β ,

$$\alpha + \beta = \frac{b}{\sqrt{2}}, \quad \alpha\beta = \frac{8-2\sqrt{5}}{\sqrt{2}}$$

The harmonic mean of the roots,

$$H = \frac{2\alpha\beta}{\alpha+\beta} = 4$$

$$\alpha\beta = 2(\alpha + \beta)$$

$$\frac{8-2\sqrt{5}}{\sqrt{2}} = 2 \cdot \left(\frac{b}{\sqrt{2}} \right)$$

$$2b = 2(4 - \sqrt{5})$$

$$b = 4 - \sqrt{5}$$

Question34

All the values of k such that the quadratic expression $2kx^2 - (4k + 1)x + 2$ is negative for exactly three integral values of x , lie in the interval

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Options:

A.

$$\left[\frac{1}{12}, \frac{1}{10} \right)$$

B.

$$\left(\frac{1}{6}, \frac{1}{5} \right)$$

C.

$$[-1, 2)$$

D.

$$[2, 6)$$

Answer: A

Solution:

We have,

$$\begin{aligned}2kx^2 - (4k + 1)x + 2 &< 0 \\ \therefore 2kx^2 - 4kx - x + 2 &< 0 \\ 2kx(x - 2) - 1(x - 2) &< 0 \\ (x - 2)(2kx - 1) &< 0 \\ 2 < x < \frac{1}{2k}\end{aligned}$$

Expression is negative for exactly 3 integral values of x which is $x = 3, 4, 5$

$$\begin{aligned}5 < x &\leq 6 \\ \therefore \frac{1}{2k} &\leq 6 \Rightarrow k \geq \frac{1}{12} \\ \therefore \frac{1}{2k} &> 5 \Rightarrow k < \frac{1}{10} \\ \therefore k &\in \left[\frac{1}{12}, \frac{1}{10} \right)\end{aligned}$$

Question35

If α, β and γ are the roots of the equation $x^3 - 13x^2 + kx + 189 = 0$ such that $\beta - \gamma = 2$, then $\beta + \gamma : k + \alpha =$

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Options:

A.

4 : 3

B.

2 : 1

C.

6 : 5

D.

3 : 4

Answer: A

Solution:

Given, $x^3 - 13x^2 + kx + 189 = 0$

$$\alpha + \beta + \gamma = 13, \alpha\beta + \beta\gamma + \gamma\alpha = k$$

$$\text{and } \alpha\beta\gamma = -189 \text{ and } \beta - \gamma = 2$$

$$[\because \gamma = \beta - 2]$$

$$\alpha + \beta + \beta - 2 = 13$$

$$\Rightarrow (15 - 2\beta)\beta(\beta - 2) = -189$$

$$\Rightarrow \beta(15\beta - 2\beta^2 - 30 + 4\beta) = -189$$

$$\Rightarrow -2\beta^3 + 19\beta^2 - 30\beta + 189 = 0$$

$$\Rightarrow 2\beta^3 - 19\beta^2 + 30\beta - 189 = 0$$

$$\Rightarrow (\beta - 9)(2\beta^2 - \beta + 21) = 0$$

$$\Rightarrow \beta = 9, 2\beta^2 - \beta + 21 \neq 0 \forall \beta \in R$$

$$\therefore \beta = 9, \gamma = 7$$

$$\text{and } \alpha = -3$$

$$k = \alpha\beta + \beta\gamma + \gamma\alpha = -27 + 63 - 21 = 15$$

$$\text{Now, } \frac{\beta + \gamma}{k + \alpha} = \frac{9 + 7}{15 - 3} = \frac{16}{12} = \frac{4}{3}$$

$$\therefore \beta + \gamma : k + \alpha = 4 : 3$$

Question36

The set of all real values of x satisfying the inequality $\frac{7x^2 - 5x - 18}{2x^2 + x - 6} < 2$ is

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Options:

A. $(-\infty, -\frac{2}{3}] \cup [3, \infty)$

B. $(-2, -\frac{2}{3}) \cup (\frac{3}{2}, 3)$

C. $(-\infty, -2) \cup (\frac{3}{2}, \infty)$

D. $[-\frac{2}{3}, \frac{3}{2})$

Answer: B

Solution:

We have, $\frac{7x^2 - 5x - 18}{2x^2 + x - 6} < 2$

$$\frac{7x^2 - 5x - 18}{2x^2 + x - 6} - 2 < 0$$

$$\frac{7x^2 - 5x - 18 - 4x^2 - 2x + 12}{2x^2 + x - 6} < 0$$

$$\frac{3x^2 - 7x - 6}{2x^2 + x - 6} < 0$$

$$\frac{3x^2 - 9x + 2x - 6}{2x^2 + 4x - 3x - 6} < 0$$

$$\frac{(3x + 2)(x - 3)}{(2x - 3)(x + 2)} < 0$$

Case I

$$(3x + 2)(x - 3) < 0 \text{ and}$$

$$(2x - 3)(x + 2) > 0$$

$$\Rightarrow x \in \left(-\frac{2}{3}, 3\right) \text{ and } x \in (-\infty, -2) \cup \left(\frac{3}{2}, 0\right)$$

$$\Rightarrow x \in \left(\frac{3}{2}, 3\right)$$

Case II

$$(3x + 2)(x - 3) > 0$$

$$\text{and } (2x - 3)(x + 2) < 0$$

$$\Rightarrow x \in \left(-\infty, -\frac{2}{3}\right) \cup (3, \infty) \text{ and}$$

$$\Rightarrow x \in \left(-2, \frac{3}{2}\right)$$

$$\Rightarrow x \in \left(-2, -\frac{2}{3}\right)$$

$$\text{Hence, } x \in \left(-2, -\frac{2}{3}\right) \cup \left(\frac{3}{2}, 3\right)$$

Question37

The set of all values of k for which the inequality $x^2 - (3k + 1)x + 4k^2 + 3k - 3 > 0$ is true for all real values of x , is

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Options:

A. $\left(-\frac{13}{7}, 1\right)$

B. $\left(-1, \frac{13}{7}\right)$

C. $\left(-\infty, -\frac{13}{7}\right) \cup (1, \infty)$

$$D. (-\infty, -1) \cup \left(\frac{13}{7}, \infty\right)$$

Answer: C

Solution:

To determine the set of all values of k for which the inequality $x^2 - (3k + 1)x + 4k^2 + 3k - 3 > 0$ holds for all real values of x , we proceed as follows.

First, recall that the inequality $f(x) > 0$ for all x requires that the quadratic equation has no real roots, which corresponds to the discriminant D being less than zero. For a quadratic equation $ax^2 + bx + c$, the discriminant is given by $b^2 - 4ac$.

In this scenario, the quadratic in x is given by:

$$a = 1, \quad b = -(3k + 1), \quad c = 4k^2 + 3k - 3.$$

The discriminant D is:

$$D = (3k + 1)^2 - 4(1)(4k^2 + 3k - 3).$$

Simplifying, we get:

$$D = (3k + 1)^2 - 16k^2 - 12k + 12.$$

Expanding:

$$D = 9k^2 + 6k + 1 - 16k^2 - 12k + 12.$$

Simplifying further:

$$D = -7k^2 - 6k + 13.$$

To ensure $D < 0$, we need:

$$7k^2 + 6k - 13 > 0.$$

To solve this inequality, factor the quadratic expression:

$$7k^2 + 13k - 7k - 13 = (7k + 13)(k - 1).$$

The solution to $(7k + 13)(k - 1) > 0$ can be found using sign analysis. The critical points are $k = -\frac{13}{7}$ and $k = 1$.

The intervals to consider are:

$$k < -\frac{13}{7}$$

$$-\frac{13}{7} < k < 1$$

$$k > 1$$

Test each interval:

For $k < -\frac{13}{7}$ and $k > 1$, both factors $(7k + 13)$ and $(k - 1)$ have the same sign (either both positive or both negative), making the product positive.

For $-\frac{13}{7} < k < 1$, the factors have opposite signs, making the product negative.

Thus, the solution set for k is:

$$k \in \left(-\infty, -\frac{13}{7}\right) \cup (1, \infty).$$

Question 38

The cubic equation whose roots are the square of the roots of the equation is

$$12x^3 - 20x^2 + x + 3 = 0$$

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Options:

A. $x^3 + 376x^2 - 121x - 9 = 0$

B. $144x^3 - 400x^2 + 121x + 98 = 0$

C. $144x^3 - 376x^2 + 121x - 9 = 0$

D. $x^3 + 400x^2 - 121x - 98 = 0$

Answer: C

Solution:

To find the cubic equation whose roots are the squares of the roots of the given equation, we start by analyzing the equation:

$$12x^3 - 20x^2 + x + 3 = 0$$

Let's consider α , β , and γ as the roots of this equation. From the properties of polynomial roots, we have:

$$\alpha + \beta + \gamma = \frac{20}{12} = \frac{5}{3}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{1}{12}$$

$$\alpha\beta\gamma = -\frac{3}{12} = -\frac{1}{4}$$

Next, we need to find relationships involving the squares of these roots α^2 , β^2 , and γ^2 .

Sum of squares of the roots:

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

$$= \left(\frac{5}{3}\right)^2 - 2 \times \frac{1}{12} = \frac{25}{9} - \frac{1}{6} = \frac{50-3}{18} = \frac{47}{18}$$

Product of squares of the roots:

$$\alpha^2\beta^2\gamma^2 = (\alpha\beta\gamma)^2 = \left(-\frac{1}{4}\right)^2 = \frac{1}{16}$$

Sum of products of squares of roots taken two at a time:

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 = (\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

$$= \left(\frac{1}{12}\right)^2 - 2 \times \left(-\frac{1}{4}\right) \times \frac{5}{3}$$

$$= \frac{1}{144} + \frac{5}{6} = \frac{1+120}{144} = \frac{121}{144}$$

Using these calculated values, the cubic equation whose roots are the squares of the original roots is given by:

$$x^3 - \frac{47}{18}x^2 + \frac{121}{144}x - \frac{1}{16} = 0$$

Multiplying through by 144 to clear fractions, we obtain:

$$144x^3 - 376x^2 + 121x - 9 = 0$$

Question39

α, β and γ are the roots of the equation $x^3 + 3x^2 - 10x - 24 = 0$ If $\alpha(\beta + \gamma), \beta(\gamma + \alpha)$ and $\gamma(\alpha + \beta)$ are the roots of the equation $x^3 + px^2 + qx + r = 0$, then q is equal to

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Options:

A. -44

B. -28

C. 44

D. 28

Answer: D

Solution:

Let α, β and γ be the roots of equation $x^3 + 3x^2 - 10x - 24 = 0$

$$\alpha + \beta + \gamma = -3, \alpha\beta + \beta\gamma + \gamma\alpha = -10$$

$$\text{and } \alpha\beta\gamma = 24$$

$$\Rightarrow \alpha = -2, \beta = -4, \gamma = 3$$

As, $\alpha(\beta + \gamma)$, $\beta(\gamma + \alpha)$ and $\gamma(\alpha + \beta)$ are the roots of equation

$$x^3 + px^2 + qx + r = 0$$

$$\begin{aligned} -p &= \alpha(\beta + \gamma) + \beta(\gamma + \alpha) + \gamma(\alpha + \beta) \\ &= -2(-4 + 3) - 4(-2 + 3) + 3(-2 - 4) \\ &= 2 - 4 - 18 \Rightarrow -20 \end{aligned}$$

$$\therefore p = 20$$

$$\begin{aligned} q &= 2(-4) + (-4)(-18) + (-18)2 \\ &= -8 + 72 - 36 = 28 \end{aligned}$$

Question40

If ' a ' is a rational number, then the roots of the equation $x^2 - 3ax + a^2 - 2a - 4 = 0$ are

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Options:

- A. rational and equal numbers
- B. different real numbers
- C. different rational numbers only
- D. not real numbers

Answer: B

Solution:

Given,

$$x^2 - 3ax + a^2 - 2a - 4 = 0$$

where, a is a rational number

Then,

discriminant, $D = B^2 - 4AC$

$$\begin{aligned} &= (-3a)^2 - 4 \times 1 \times (a^2 - 2a - 4) \\ &= 9a^2 - 4a^2 + 8a + 16 \\ &= 5a^2 + 8a + 16 \\ D &= \underbrace{5a^2 + 16 + 8a}_{+ve} \end{aligned}$$

So, $D > 0$

So, roots of equation will be different real number.

Question41

The set of all real values ' a ' for which $-1 < \frac{2x^2+ax+2}{x^2+x+1} < 3$ holds for all real values of x is

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Options:

A. $(-7, 5)$

B. $(5, \infty)$

C. $(1, 5)$

D. $(-\infty, 1)$

Answer: C

Solution:

$$\text{Given, } -1 < \frac{2x^2 + ax + 2}{x^2 + x + 1} < 3$$

$$\text{Now, } -1 < \frac{2x^2 + ax + 2}{x^2 + x + 1}$$

$$\Rightarrow -x^2 - x - 1 < 2x^2 + ax + 2$$

$$\Rightarrow 0 < 3x^2 + (a+1)x + 3$$

$$\Rightarrow D < 0$$

$$\Rightarrow (a+1)^2 - 4 \times 3 \times 3 < 0$$

$$\Rightarrow (a+1)^2 - 36 < 0 \Rightarrow (a+1)^2 - 6^2 < 0$$

$$\Rightarrow (a+1-6)(a+1+6) < 0$$

$$\Rightarrow (a-5)(a+7) < 0$$

$$\text{and } \frac{2x^2 + ax + 2}{x^2 + x + 1} < 3 \quad \dots (i)$$

$$\Rightarrow 2x^2 + ax + 2 < 3x^2 + 3x + 3$$

$$\Rightarrow 0 < x^2 + (3-a)x + 1 \Rightarrow D < 0$$

$$\Rightarrow (3-a)^2 - 4 \times 1 \times 1 < 0$$

$$\Rightarrow (3-a)^2 - 2^2 < 0$$

$$\Rightarrow (3-a-2)(3-a+2) < 0$$

$$\Rightarrow (-a+1)(5-a) < 0$$

$\Rightarrow (a - 5)(-1 + a) < 0$... (ii)
From Eq. (i) and (ii), we get
 $a \in (1, 5)$

Question42

The quotient, when $3x^5 - 4x^4 + 5x^3 - 3x^2 + 6x - 8$ is divided by $x^2 + x - 3$ is

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Options:

A. $3x^2 - 7x - 21$

B. $3x^3 - 7x^2 + 21x - 45$

C. $3x^4 - 7x^3 + 21x^2 - 45 + 114$

D. $114x - 143$

Answer: B

Solution:

$$\begin{array}{l} x^2 + x - 3) \\ 3x^5 - 4x^4 + 5x^3 - 3x^2 + 6x - 8 \\ (3x^3 - 7x^2 + 21x - 45 \end{array}$$

$$\begin{array}{r}
3x^5 + 3x^4 - 9x^3 \\
- \quad - \quad + \\
\hline
-7x^4 + 14x^3 - 3x^2 + 6x - 8 \\
-7x^4 - 7x^3 + 21x^2 \\
+ \quad + \quad - \\
\hline
21x^3 - 24x^2 + 6x - 8 \\
21x^3 - 21x^2 + 63x \\
- \quad + \quad + \\
\hline
-45x^2 + 69x - 8 \\
-45x^2 + 45x - 135 \\
+ \quad + \quad + \\
\hline
114x + 127 \\
\hline
\end{array}$$

Hence, quotient is $3x^3 - 7x^2 + 21x - 45$

Question43

If both the roots of the equation $x^2 - 6ax + 2 - 2a + 9a^2 = 0$ exceed 3, then

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Options:

A. $a < \frac{3}{2}$

B. $a > \frac{3}{2}$

C. $a < \frac{5}{2}$

D. $a > \frac{11}{9}$

Answer: D

Solution:

Given, $x^2 - 6ax + 2 - 2a + 9a^2 = 0$

$\Rightarrow x^2 - 6ax + (2 - 2a + 9a^2) = 0$

For both roots to exceed 3, $x = 3$ should be a root or both roots should be greater than 3.

$$\text{Sum of roots} = -\frac{b}{a} = -\left(\frac{-6a}{1}\right) = 6a$$

$$6a > 18 \Rightarrow a > 3$$

$$\text{Product of roots} = \frac{c}{a} = \frac{9a^2 - 2a + 2}{1}$$

Since, $a > 3$ so product, also should be greater than 3 .

That leads us to the correct condition

$$a > \frac{11}{9}$$

Question44

If α and β are two distinct negative roots of $x^5 - 5x^3 + 5x^2 - 1 = 0$, then the equation of least degree with integer coefficients having $\sqrt{-\alpha}$ and $\sqrt{-\beta}$ as its roots, is

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Options:

A. $x^2 - 3x + 1 = 0$

B. $-x^4 - 5x^2 + 5x + 1 = 0$

C. $-x^4 + 5x^2 - 5x + 1 = 0$

D. $x^4 - 3x^2 + 1 = 0$

Answer: D

Solution:

We begin with the polynomial:

$$x^5 - 5x^3 + 5x^2 - 1 = (x - 1)^3(x^2 + 3x + 1)$$

We are given that α and β are distinct negative roots of the equation. Therefore, α and β must be the roots of the quadratic equation:

$$x^2 + 3x + 1 = 0$$

Given that α and β are negative, we establish that:

$$\alpha\beta = 1 \quad (\text{as derived from the product of the roots of } x^2 + 3x + 1 = 0)$$

We need to find the polynomial equation for which $\sqrt{-\alpha}$ and $\sqrt{-\beta}$ are roots. Note:

$$\sqrt{-\alpha} \cdot \sqrt{-\beta} = \sqrt{\alpha \cdot \beta} = \sqrt{1} = 1$$

Thus, the product of $\sqrt{-\alpha}$ and $\sqrt{-\beta}$ is 1. The equation with these roots and integer coefficients is:

$$x^4 - 3x^2 + 1 = 0$$

The product of the roots for this equation equals 1, satisfying the condition $\sqrt{-\alpha} \cdot \sqrt{-\beta} = 1$, confirming that $\sqrt{-\alpha}$ and $\sqrt{-\beta}$ are roots of the equation.

Question45

If α is a common root of $x^2 - 5x + \lambda = 0$ and $x^2 - 8x - 2\lambda = 0 (\lambda \neq 0)$ and β, γ are the other roots of them, then $\alpha + \beta + \gamma + \lambda =$

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Options:

A. 0

B. -1

C. 1

D. 2

Answer: C

Solution:

Given,

$$x^2 - 5x + \lambda = 0 \quad \dots (i)$$

$$x^2 - 8x - 2\lambda = 0 \quad \dots (ii)$$

$\therefore \alpha$ is a common factor of Eqs (i) and (ii), we get

And β, γ are the other roots of them,

So

$$\alpha^2 - 5\alpha + \lambda = 0$$

$$\alpha^2 - 8\alpha - 2\lambda = 0$$

$$\begin{array}{r} -(+)\quad (+) \\ \hline 3\alpha + 3\lambda = 0 \Rightarrow \lambda = -\alpha \end{array}$$

$$\text{Then, } \alpha^2 - 5\alpha - \alpha = 0 \Rightarrow \alpha^2 - 6\alpha = 0$$

$$\Rightarrow \alpha(\alpha - 6) = 0 \Rightarrow \alpha = 0, 6$$

$$\alpha \neq 0$$

$$\Rightarrow \alpha = 6 \text{ and } \lambda = -6$$

Now, from Eq. (i), we get

$$x^2 - 5x - 6 = 0$$

$$\Rightarrow x^2 - 6x + x - 6 = 0$$

$$\Rightarrow (x - 6)(x + 1) = 0$$

$$\Rightarrow x = 6, x = -1$$

$$\therefore \beta = -1$$

From Eq. (ii), we get

$$x^2 - 8x - 2 \times (-6) = 0$$

$$\Rightarrow x^2 - 8x + 12 = 0$$

$$\Rightarrow (x - 6)(x - 2) = 0 \Rightarrow x = 6, 2$$

$$\therefore \gamma = 2$$

$$\text{Now, } \alpha + \beta + \gamma + \lambda = 6 + (-1) + 2 + (-\theta)$$

$$= 1$$

Question46

The equation $x^4 - x^3 - 6x^2 + 4x + 8 = 0$ has two equal roots. If α, β are the other two roots of this equation, then $\alpha^2 + \beta^2 =$

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Options:

A. 4

B. 5

C. 6

D. 7

Answer: B

Solution:

To solve the given polynomial equation $x^4 - x^3 - 6x^2 + 4x + 8 = 0$ with two equal roots, we proceed as follows:

The equation can be expressed as:

$$x^3(x+1) - 2x^2(x+1) - 4x(x+1) + 8(x+1) = 0$$

Factoring out $(x+1)$ gives:

$$(x+1)(x^3 - 2x^2 - 4x + 8) = 0$$

Further factoring results in:

$$(x+1)(x-2)(x^2 - 4) = 0$$

This simplifies to:

$$x+1=0, \quad x-2=0, \quad x^2-4=0$$

Hence, the roots are:

$$x = -1, \quad x = 2, \quad x = \pm 2$$

The distinct roots of the equation are -2 , -1 , and 2 . Since one root, 2 , repeats, we determine the remaining roots as $\alpha = -2$ and $\beta = -1$.

To find $\alpha^2 + \beta^2$, compute:

$$\alpha^2 + \beta^2 = (-2)^2 + (-1)^2 = 4 + 1 = 5$$

Thus, $\alpha^2 + \beta^2 = 5$.

Question47

Roots of the equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ are

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Options:

A. $\frac{a(b-c)}{c(a-b)}, 1$

B. $\frac{b(c-a)}{c(a-b)}, 1$

C. $\frac{c(a-b)}{a(b-c)}, 1$

D. $\frac{c(a-b)}{b(c-a)}, 1$

Answer: C

Solution:

We have,

$$a(b-c)x^2 + b(c-a)x + c(a-b) = 0$$

Using hit and trial method,

Put $x = 1$ in Eq. (i)

$$\begin{aligned} a(b-c) + b(c-a) + c(a-b) &= 0 \\ ab - ac + bc - ab + ac - bc &= 0 \end{aligned}$$

$\therefore$ one root is 1 .

Now, let another root be α

$$\text{Then, product of root, } \alpha \cdot 1 = \frac{c(a-b)}{a(b-c)}$$

$$\Rightarrow \alpha = \frac{c(a-b)}{a(b-c)}$$

Hence, $\frac{c(a-b)}{a(b-c)}, 1$

Question48

The algebraic equation of degree 4 whose roots are translate of the roots of the equation. $x^4 + 5x^3 + 6x^2 + 7x + 9 = 0$ by -1 is

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Options:

A. $x^4 + x^3 - 3x^2 + 6x + 4 = 0$

B. $x^4 + 9x^3 + 27x^2 + 38x + 28 = 0$

C. $x^4 + 5x^3 + 6x^2 + 7x + 9 = 0$

D. $x^4 - 5x^3 + 6x^2 - 7x + 9 = 0$

Answer: A

Solution:

To find the algebraic equation of degree 4 whose roots are translations of the roots of the given equation $x^4 + 5x^3 + 6x^2 + 7x + 9 = 0$ by -1 , we perform a substitution where x becomes $x - 1$.

Given the equation:

$$x^4 + 5x^3 + 6x^2 + 7x + 9 = 0,$$

we translate the roots by -1 , meaning we consider $x = t - 1$, which implies $t = x + 1$.

We need to find $f(t - 1) = 0$:

$$\begin{aligned}
 f(t-1) &= (t-1)^4 + 5(t-1)^3 + 6(t-1)^2 + 7(t-1) + 9 \\
 &= (t^4 - 4t^3 + 6t^2 - 4t + 1) \\
 &\quad + 5(t^3 - 3t^2 + 3t - 1) \\
 &\quad + 6(t^2 - 2t + 1) \\
 &\quad + 7(t-1) + 9.
 \end{aligned}$$

Simplifying each expression:

$$(t^4 - 4t^3 + 6t^2 - 4t + 1)$$

$$5(t^3 - 3t^2 + 3t - 1) = 5t^3 - 15t^2 + 15t - 5$$

$$6(t^2 - 2t + 1) = 6t^2 - 12t + 6$$

$$7(t-1) = 7t - 7$$

Adding these terms together:

$$\begin{aligned}
 &t^4 - 4t^3 + 6t^2 - 4t + 1 \\
 &+ 5t^3 - 15t^2 + 15t - 5 \\
 &+ 6t^2 - 12t + 6 \\
 &+ 7t - 7 + 9 = 0
 \end{aligned}$$

Combine like terms:

$$\begin{aligned}
 &t^4 + (5t^3 - 4t^3) + (6t^2 - 15t^2 + 6t^2) + (-4t + 15t - 12t + 7t) + (1 - 5 + 6 - 7 + 9) \\
 &= t^4 + t^3 - 3t^2 + 6t + 4 = 0.
 \end{aligned}$$

Thus, the required equation is:

$$x^4 + x^3 - 3x^2 + 6x + 4 = 0.$$

Question49

Let $[r]$ denote the largest integer not exceeding r and the roots of the equation $3x^2 + 6x + 5 + \alpha(x^2 + 2x + 2) = 0$ are complex number when ever $\alpha > L$ and α

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Options:

A. L

B. M

C. $L + M$

D. $M - L$

Answer: A

Solution:

Given equation is

$$3x^2 + 6x + 5 + \alpha(x^2 + 2x + 2) = 0$$
$$\Rightarrow (3 + \alpha)x^2 + (6 + 2\alpha)x + (5 + 2\alpha) = 0$$

Now, discriminants,

$$b^2 - 4ac = (6 + 2\alpha)^2 - 4(3 + \alpha)(5 + 2\alpha)$$
$$= 36 + 4\alpha^2 + 24\alpha - 60 - 44\alpha - 8\alpha^2$$
$$= -4\alpha^2 - 20\alpha - 24$$

Since, root are complex, so, discriminant is negative

$$-4\alpha^2 - 20\alpha - 24 < 0$$
$$\Rightarrow \alpha^2 + 5\alpha + 6 > 0$$
$$\Rightarrow (\alpha + 2)(\alpha + 3) > 0$$
$$\Rightarrow \alpha < -3 \text{ and } \alpha > -2$$

On comparing, we get

$$M = -3, L = -2$$

$$\text{Now, } L - M = -2 - (-3) = -2 + 3 = 1$$

$$\text{Now, } Ly^2 + My + r < 0$$

$$\Rightarrow -2y^2 - 3y + r < 0$$

$$\Rightarrow 2y^2 + 3y - r > 0$$

$$\text{So, } 9 + 8r < 0 = 0 \Rightarrow r < -9/8$$

$$\text{Hence, } [r] = -2 = L$$

Question50

For any real value of x . If $\frac{11x^2+12x+6}{x^2+4x+2} \notin (a, b)$, then the value x for which $\frac{11x^2+12x+6}{x^2+4x+2} = b - a + 3$ is

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Options:

A. $\frac{3}{4}$

B. $\frac{3}{2}$

C. 2

D. $-\frac{1}{2}$

Answer: D

Solution:

To solve the given problem, we start by defining the expression:

$$y = \frac{11x^2 + 12x + 6}{x^2 + 4x + 2}$$

This can be rewritten as:

$$y(x^2 + 4x + 2) = 11x^2 + 12x + 6$$

Rearranging the equation, we get:

$$(y - 11)x^2 + (4y - 12)x + (2y - 6) = 0$$

We will analyze this by considering two cases:

Case I: $y \neq 11$

For this quadratic equation in x , the discriminant D must be non-negative for real solutions:

$$D = (4y - 12)^2 - 4(y - 11)(2y - 6) \geq 0$$

Expanding this, we get:

$$\begin{aligned} D &= (4y - 12)^2 - 4(y - 11)(2y - 6) \\ &= 16y^2 - 96y + 144 - 8y^2 + 44y - 264 \\ &= 8y^2 - 52y - 120. \end{aligned}$$

Simplifying, we have:

$$y^2 + 2y - 15 \geq 0.$$

Factorizing, we find:

$$(y + 5)(y - 3) \geq 0.$$

This inequality implies:

$$y \leq -5 \quad \text{or} \quad y \geq 3.$$

Case II: $y = 11$

If $y = 11$, the quadratic equation becomes linear, and from the simplified version:

$$(44y - 12)x + 22y - 6 = 0$$

Simplifying gives:

$$(32)x + 16 = 0$$

Solving for x , we get:

$$x = -\frac{1}{2}.$$

Substituting $x = -\frac{1}{2}$ back into the expression for y :

$$y = \frac{11(-\frac{1}{2})^2 + 12(-\frac{1}{2}) + 6}{(-\frac{1}{2})^2 + 4(-\frac{1}{2}) + 2} = 11 \notin (-5, 3).$$

Thus, the value of x for which $y = b - a + 3$ is:

$$x = -\frac{1}{2}.$$

Question 51

If the roots of $\sqrt{\frac{1-y}{y}} + \sqrt{\frac{y}{1-y}} = \frac{5}{2}$ are α and β ($\beta > \alpha$) and the equation $(\alpha + \beta)x^4 - 25\alpha\beta x^2 + (\gamma + \beta - \alpha) = 0$ has real roots, then a possible value of γ is

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Options:

A. $\frac{1}{2}$

B. 4

C. 2π

D. $\sqrt{e+13}$

Answer: A

Solution:

To solve the problem, we begin by analyzing the equation:

$$\frac{\sqrt{1-y}}{\sqrt{y}} + \frac{\sqrt{y}}{\sqrt{1-y}} = \frac{5}{2}$$

This simplifies to:

$$\frac{1-y+y}{\sqrt{y(1-y)}} = \frac{5}{2}$$

Hence,

$$\frac{1}{\sqrt{y(1-y)}} = \frac{5}{2}$$

Squaring both sides gives:

$$4 = 25y(1 - y)$$

Simplifying, we obtain:

$$25y^2 - 25y + 4 = 0$$

Solving this quadratic equation yields:

$$y = \frac{4}{5}, \frac{1}{5}$$

Thus, the roots are $\alpha = \frac{1}{5}$ and $\beta = \frac{4}{5}$ with $\beta > \alpha$.

Next, consider the given equation:

$$(\alpha + \beta)x^4 - 25\alpha\beta x^2 + (\gamma + \beta - \alpha) = 0$$

Substituting $\alpha = \frac{1}{5}$ and $\beta = \frac{4}{5}$, we have:

$$x^4 - 4x^2 + \left(\gamma + \frac{3}{5}\right) = 0$$

For this equation to have real roots, the discriminant D of the quadratic x^2 term must be non-negative:

$$D = 16 - 4\left(\gamma + \frac{3}{5}\right) \geq 0$$

Solving for γ , we find:

$$4\left(\gamma + \frac{3}{5}\right) \leq 16$$

$$\gamma \leq 4 - \frac{3}{5} = \frac{17}{5} = 3.4$$

Thus, a possible value of γ is $\frac{1}{2}$.

Question52

If α and β are two double roots of $x^2 + 3(a + 3)x - 9a = 0$ for different values of a ($\alpha > \beta$), then the minimum value of $x^2 + \alpha x - \beta = 0$ is

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Options:

A. $\frac{69}{4}$

B. $-\frac{69}{4}$

C. $-\frac{35}{4}$

D. $\frac{35}{4}$

Answer: B

Solution:

To solve the problem, we start with the given quadratic equation where α and β are double roots for different values of a :

$$x^2 + 3(a + 3)x - 9a = 0.$$

For the roots to be double, the discriminant D must equal zero:

$$D = (3(a + 3))^2 - 4 \times 1 \times (-9a) = 0.$$

This simplifies to:

$$9(a + 3)^2 + 36a = 0.$$

Expanding and simplifying, we get:

$$9(a^2 + 6a + 9) + 36a = 0,$$

$$9a^2 + 54a + 81 + 36a = 0,$$

$$9a^2 + 90a + 81 = 0.$$

Divide the entire equation by 9:

$$a^2 + 10a + 9 = 0.$$

Solving this quadratic equation:

$$a = \frac{-10 \pm \sqrt{100 - 36}}{2} = \frac{-10 \pm \sqrt{64}}{2}.$$

$$a = \frac{-10 \pm 8}{2}.$$

This results in $a = -1$ and $a = -9$.

For $a = -9$:

The quadratic becomes:

$$x^2 + 3(-9 + 3)x + 81 = 0 \Rightarrow x^2 - 18x + 81 = 0.$$

The double root is:

$$x = 9.$$

For $a = -1$:

The quadratic becomes:

$$x^2 + 3(2)x + 9 = 0 \Rightarrow x^2 + 6x + 9 = 0.$$

The double root is:

$$x = -3.$$

From this, we identify $\alpha = 9$ and $\beta = -3$.

Now consider the equation:

$$x^2 + \alpha x - \beta = 0 \Rightarrow x^2 + 9x + 3 = 0.$$

The minimum value of a quadratic equation $ax^2 + bx + c$ is given by:

$$c - \frac{b^2}{4a}.$$

Plugging in the coefficients:

$$= 3 - \frac{81}{4} = -\frac{69}{4}.$$

Question53

If $2x^2 + 3x - 2 = 0$ and $3x^2 + \alpha x - 2 = 0$ have one common root, then the sum of all possible values of α is

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Options:

A. -35

B. 7.5

C. -7.5

D. -1.5

Answer: B

Solution:

To solve the problem, we have two quadratic equations, one of which is:

$$2x^2 + 3x - 2 = 0$$

First, we find the roots of this equation using the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Substituting the specific values:

$$x = \frac{-3 \pm \sqrt{3^2 - 4 \times 2 \times (-2)}}{2 \times 2}$$

$$x = \frac{-3 \pm \sqrt{9 + 16}}{4}$$

$$x = \frac{-3 \pm \sqrt{25}}{4}$$

$$x = \frac{-3 \pm 5}{4}$$

Thus, the roots of the first equation are:

$$x = \frac{2}{4}, \frac{-8}{4} = \frac{1}{2}, -2$$

Given that the second equation:

$$3x^2 + \alpha x - 2 = 0$$

shares one root with the first equation, we need to substitute these roots into the second equation.

Case 1: Root $x = \frac{1}{2}$

Substitute into the second equation:

$$3\left(\frac{1}{2}\right)^2 + \alpha\left(\frac{1}{2}\right) - 2 = 0$$

$$3 \times \frac{1}{4} + \frac{1}{2}\alpha - 2 = 0$$

$$\frac{3}{4} + \frac{\alpha}{2} - 2 = 0$$

Solving for α :

$$\frac{\alpha}{2} = 2 - \frac{3}{4}$$

$$\frac{\alpha}{2} = \frac{5}{4}$$

$$\alpha = \frac{5}{2}$$

Case 2: Root $x = -2$

Substitute into the second equation:

$$3(-2)^2 + \alpha(-2) - 2 = 0$$

$$12 - 2\alpha - 2 = 0$$

$$2\alpha = 10$$

$$\alpha = 5$$

Sum of All Possible Values of α

Adding the solutions:

$$\alpha = \frac{5}{2} + 5 = \frac{5}{2} + \frac{10}{2} = \frac{15}{2} = 7.5$$

Thus, the sum of all possible values of α is 7.5.

Question 54

If the sum of two roots of $x^3 + px^2 + qx - 5 = 0$ is equal to its third root, then $p(p^2 - 4q) =$

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Options:

A. -20

B. 20

C. 40

D. -40

Answer: C**Solution:**

Let α, β, γ be the roots of the equation $x^3 + px^2 + qx - 5 = 0$.

Given that $\alpha + \beta = \gamma$, we can derive the following:

From Vieta's formulas, we know:

$$\alpha + \beta + \gamma = -p$$

Substituting the given condition $\alpha + \beta = \gamma$ into the equation, we have:

$$\alpha + \beta + \gamma = \gamma + \gamma = 2\gamma,$$

$$2\gamma = -p,$$

$$\gamma = \frac{-p}{2}.$$

Another Vieta formula gives us:

$$\alpha\beta + \beta\gamma + \gamma\alpha = q.$$

Substituting for $\alpha + \beta = \gamma$, we get:

$$\alpha\beta + \gamma(\beta + \alpha) = q,$$

$$\alpha\beta + \gamma^2 = q,$$

$$\alpha\beta + \frac{p^2}{4} = q.$$

$$\text{Thus, } \alpha\beta = q - \frac{p^2}{4}.$$

Using the product of roots from Vieta's formula:

$$\alpha\beta\gamma = 5.$$

Substituting $\gamma = \frac{-p}{2}$ and $\alpha\beta = q - \frac{p^2}{4}$, we have:

$$\begin{aligned}\left(q - \frac{p^2}{4}\right) \left(-\frac{p}{2}\right) &= 5, \\ \left(\frac{4q - p^2}{4}\right) \left(-\frac{p}{2}\right) &= 5, \\ \frac{-p(4q - p^2)}{8} &= 5, \\ p(4q - p^2) &= -40, \\ p(p^2 - 4q) &= 40.\end{aligned}$$

Thus, the value is 40.

Question55

$$4 + \frac{1}{4 + \frac{1}{4 + \frac{1}{4 + \dots \infty}}} =$$

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Options:

A. $(2 + \sqrt{5}), (2 - \sqrt{5})$

B. $2 + \sqrt{5}$

C. $2 - \sqrt{5}$

D. $2 + \sqrt{3}$

Answer: B

Solution:

$$\begin{aligned}\text{Let } x &= 4 + \frac{1}{4 + \frac{1}{4 + \frac{1}{4 + \dots \infty}}} \\ \Rightarrow x &= 4 + \frac{1}{x} \\ \Rightarrow x^2 - 4x - 1 &= 0 \\ \Rightarrow x &= \frac{4 \pm \sqrt{16 + 4}}{2} = 2 \pm \sqrt{5}\end{aligned}$$

As the value of x is greater than 4, $x = 2 + \sqrt{5}$

Question56

If $x^2 + 5ax + 6 = 0$ and $x^2 + 3ax + 2 = 0$ have a common root, then that common root is

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Options:

A. 3 (or) -3

B. 2 (or) -2

C. 2 (or) -3

D. -2 (or) 3

Answer: B

Solution:

Let α be the common root

$$\Rightarrow \alpha^2 + 5a\alpha + 6 = 0$$

$$\text{and } \alpha^2 + 3a\alpha + 2 = 0$$

On subtracting Eq. (ii) from Eq. (i), we get $-2a\alpha - 4 = 0 \Rightarrow a\alpha = -2$ On Putting the value of $a\alpha = -2$ in Eq. (i)

$$\alpha^2 - 10 + 6 = 0 \Rightarrow \alpha^2 = 4 \Rightarrow \alpha = \pm 2$$

Hence, the common roots is 2 or -2 .

Question57

If α, β, γ are roots of equations $x^3 + ax^2 + bx + x = 0$, then $\alpha^{-1} + \beta^{-1} + \gamma^{-1} =$

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Options:

A. $\frac{a}{c}$

B. $-\frac{b}{c}$

C. $\frac{c}{a}$

D. $\frac{b}{a}$

Answer: B

Solution:

To find $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$, we consider that α , β , and γ are the roots of the equation:

$$x^3 + ax^2 + bx + c = 0.$$

From Vieta's formulas, we know:

$$\alpha + \beta + \gamma = -a,$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = b,$$

$$\alpha\beta\gamma = -c.$$

To find the sum of the reciprocals of the roots:

$$\alpha^{-1} + \beta^{-1} + \gamma^{-1} = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

Utilizing the identity for sums of reciprocals, this can be rewritten as:

$$\alpha^{-1} + \beta^{-1} + \gamma^{-1} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}.$$

Substituting the known values from Vieta's formulas, we get:

$$\alpha^{-1} + \beta^{-1} + \gamma^{-1} = \frac{b}{-c} = -\frac{b}{c}.$$

Question58

For all positive integers n if $3^{2n+1} + 2^{n+1}$ is divisible by k , then the number of prime numbers less than or equal to k is

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Options:

A. 17

B. 6

C. 7

D. 8

Answer: C

Solution:

Given the expression $P(n) = 3^{2n+1} + 2^{n+1}$, we know it is divisible by k for all positive integers n . To find k , let's evaluate $P(n)$ for specific values:

First, compute $P(1)$:

$$P(1) = 3^3 + 2^2 = 3 \times 27 + 4 = 81 + 4 = 85$$

Next, compute $P(2)$:

$$P(2) = 3^5 + 2^3 = 3 \times 243 + 8 = 729 + 8 = 737$$

Both $P(1)$ and $P(2)$ should be divisible by k . Now, find the greatest common divisor (GCD) of 85 and 737 to determine k :

$$k = \text{GCD}(85, 737)$$

Using the Euclidean algorithm, find the GCD:

$$737 \div 85 = 8, \text{ remainder } 57$$

$$85 \div 57 = 1, \text{ remainder } 28$$

$$57 \div 28 = 2, \text{ remainder } 1$$

$$28 \div 1 = 28, \text{ remainder } 0$$

Thus, the GCD is 1. This calculation contradicts the intended determination of k , which suggests revisiting the expression and computations for errors or adjustments.

Given the explanation involving $\text{HCF}(391, 9503) = 17$, this indicates $k = 17$.

Now, identify the prime numbers less than or equal to 17:

$$2, 3, 5, 7, 11, 13, 17$$

In total, we have 7 prime numbers. Therefore, the number of prime numbers less than or equal to k is 7.

Question59

If the roots of the quadratic equation $x^2 - 35x + c = 0$ are in the ratio $2 : 3$ and $c = 6K$, then $K =$

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Options:

A. 49

B. 14

C. 21

D. 7

Answer: A

Solution:

The roots of the quadratic equation $x^2 - 35x + c = 0$ are given to be in the ratio 2:3.

Let's assume the roots are 2α and 3α .

The sum of the roots is given by the equation:

$$2\alpha + 3\alpha = 5\alpha = 35$$

Solving for α , we get:

$$\alpha = 7$$

The product of the roots can be expressed as:

$$(2\alpha)(3\alpha) = 6\alpha^2 = c$$

Substituting the value of α , we find:

$$c = 6 \times 7^2 = 6 \times 49$$

Given that $c = 6K$, we equate to find K :

$$6 \times 49 = 6K$$

$$K = 49$$

Question60

If the sum of two roots α, β of the equation $x^4 - x^3 - 8x^2 + 2x + 12 = 0$ is zero and $\gamma, \delta (\gamma > \delta)$ are its other roots, then $3\gamma + 2\delta =$

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Options:

A. 0

B. 1

C. 3

D. 5

Answer: D

Solution:

We are given the polynomial equation $x^4 - x^3 - 8x^2 + 2x + 12 = 0$ with roots α, β, γ , and δ . It is known that the sum of two of the roots, α and β , is zero, i.e., $\alpha + \beta = 0$.

From Vieta's formulas, the sum of all roots is:

$$S_1 = \alpha + \beta + \gamma + \delta = 1$$

Given $\alpha + \beta = 0$, we find:

$$\gamma + \delta = 1 \quad \dots (ii)$$

Similarly, from Vieta's formulas, the sum of the products of the roots taken two at a time is:

$$S_2 = \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = -8$$

Using $\alpha + \beta = 0$, simplify to:

$$\alpha\beta + \gamma\delta = -8 \quad \dots (iii)$$

From the sum of the products of the roots taken three at a time, we have:

$$S_3 = \alpha\beta\gamma + \alpha\beta\delta + \alpha\gamma\delta + \beta\gamma\delta = -2$$

Substitute $(\alpha + \beta)\gamma\delta + \alpha\beta(\gamma + \delta) = -2$ and rearrange:

$$\alpha\beta = -2 \quad \dots (iv)$$

From Vieta's formulas, the product of the roots is given by:

$$S_4 = \alpha\beta\gamma\delta = 12$$

Use $\alpha\beta = -2$ to find:

$$\gamma\delta = -6 \quad \dots (v)$$

Next, solve the system of equations obtained from (ii) and (v):

$$\gamma + \delta = 1$$

$$\gamma\delta = -6$$

These are the equations for a quadratic:

$$t^2 - (\gamma + \delta)t + \gamma\delta = 0 \implies t^2 - t - 6 = 0$$

Solving, we find the roots to be:

$$t = \frac{1 \pm \sqrt{1+24}}{2} = \frac{1 \pm 5}{2}$$

Thus, $t = 3$ or $t = -2$. Since $\gamma > \delta$, we assign $\gamma = 3$ and $\delta = -2$.

Finally, compute:

$$3\gamma + 2\delta = 3 \times 3 + 2 \times (-2) = 9 - 4 = 5$$

Question61

If $S = \{m \in R : x^2 - 2(1 + 3m)x + 7(3 + 2m) = 0 \text{ has distinct roots}\}$, then the number of elements in S is

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Options:

A. 2

B. 3

C. 4

D. infinite

Answer: D

Solution:

Given, equation $x^2 - 2(1 + 3m)x + 7(3 + 2m) = 0$

Here, $a > 0$ and $D = b^2 - 4ac > 0$

Expression is always positive, if roots are distinct then $D > 0$,

$$[-2(1 + 3m)]^2 - 4 \times 7(3 + 2m) > 0$$

$$\Rightarrow 4[1 + 9m^2 + 6m] - 84 - 56m > 0$$

$$\Rightarrow 36m^2 - 32m - 80 > 0$$

$$\Rightarrow 9m^2 - 8m - 20 > 0$$

$$\Rightarrow (m - 2) \left(m + \frac{10}{9} \right) > 0$$

$$\Rightarrow m \in \left(-\infty, -\frac{10}{9} \right) \cup [2, \infty)$$

$\therefore$ Integral value of m are 3, 1, 2, 3, 4, 5, 6. So, the number of element in S is infinite.

Question62

The sum of the real roots of the equation $x^4 - 2x^3 + x - 380 = 0$ is

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Options:

A. -1

B. 0

C. 1

D. 2

Answer: C

Solution:

Considering the question to be, to find the roots of the equation $x^4 - 2x^3 + x - 380 = 0$

Now, by trial error method we find that $x = 5$ is a solution to the equation

$$\begin{aligned}\Rightarrow 5^4 - 2 \times 5^3 + 5 - 380 &= 0 \\ &= 625 - 250 + 5 - 380 = 0 \\ &= 380 - 380 = 0\end{aligned}$$

Thus, $x = 5$ is a solution.

We also find that $x = -4$ is a solution

$$\begin{aligned}(-4)^4 - 2 \times (-4)^3 - 4 - 380 &= 0 \\ &= 256 + 128 - 4 - 380 \\ &= 380 - 380 = 0\end{aligned}$$

The real roots are 5 and -4 .

Sum of real roots are $5 + (-4) = 1$

Question63

If one root of the cubic equation $x^3 + 36 = 7x^2$ is double of another, then the number of negative roots are

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Options:

A. 1

B. 2

C. 3

D. 0

Answer: A

Solution:

Let $a, 2a$ and b be roots of the cubic equation $x^3 + 36 - 7x^2 = 0$

So, sum of roots $= -\frac{(-7)}{1}$

$$\therefore a + 2a + b = 7$$

$$\Rightarrow 3a + b = 7 \quad \dots (i)$$

Now, $a(2a)b = 36$ [product of roots $= \frac{d}{a}$]

$$2a^2b = 36 \text{ and } ab + a(2a) + 2a(b) = 0$$

$$\Rightarrow 2a^2 + 3ab = 0 \Rightarrow a(2a + 3b) = 0$$

$$\Rightarrow b = \frac{-2a}{3}$$

$$\Rightarrow 3a - \frac{2a}{3} = 7 \quad [\text{from Eq. (i)}]$$

$$\Rightarrow \frac{9a - 2a}{3} = 7 \Rightarrow 9a - 2a = 21$$

$$\Rightarrow 7a = 21 \Rightarrow a = 3$$

$$\text{and } b = 7 - 3a$$

$$= 7 - 3(3) = -2 \quad [\text{from Eq. (i)}]$$

Roots are $3, 6, -2$

$\therefore$ Number of negative root is 1.

Question64

If $f(f(0)) = 0$, where $f(x) = x^2 + ax + b, b \neq 0$, then $a + b =$

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Options:

A. 2

B. 1

C. -1

D. -2

Answer: C

Solution:

$$\text{Given, } f(x) = x^2 + ax + b(b \neq 0)$$

$$\Rightarrow f(f(x)) = (x^2 + ax + b)^2 + a(x^2 + ax + b) + b$$

$$\text{Since, } f(f(0)) = 0$$

$$\text{Now, } b^2 + ab + b = 0 \dots (i)$$

Take $a = 0$ and $b = -1$, then

Eq. (i) is satisfied.

$$\text{Hence, } a + b = 0 + (-1) = -1$$

Question65

The sum of the real roots of the equation $|x - 2|^2 + |x - 2| - 2 = 0$ is

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Options:

A. 4

B. -4

C. 2

D. -2

Answer: A

Solution:

$$\text{Let } f(x) = |x - 2|$$
$$\begin{cases} x - 2, & x > 2 \\ -(x - 2), & x < 2 \end{cases}$$

Then, for $x > 2$ the equation becomes

$$\begin{aligned}\Rightarrow (x-2)^2 + (x-2) - 2 &= 0 \\ \Rightarrow x^2 - 3x &= 0 \\ \Rightarrow x &= 0, 3\end{aligned}$$

Thus, the root of the equation for $x > 2$ is 3.

For $x < 0$, the equation becomes

$$\begin{aligned}(x-2)^2 + (2-x) - 2 &= 0 \\ \Rightarrow x^2 - 5x + 4 &= 0 \\ \Rightarrow (x-4)(x-1) &= 0 \\ \Rightarrow x &= 4, 1\end{aligned}$$

The root which is less than 2 is 1.

Thus, the roots of given equations are 3, 1. Sum will be $3 + 1 = 4$

Question66

If the difference between the roots of $x^2 + ax + b = 0$ and that of the roots of $x^2 + bx + a = 0$ is same and $a \neq b$, then

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Options:

A. $a - b - 4 = 0$

B. $a - b + 4 = 0$

C. $a + b + 4 = 0$

D. $a + b - 4 = 0$

Answer: C

Solution:

Let α_1 and α_2 be roots of the equation $x^2 + ax + b = 0$.

Therefore, $\alpha_1 + \alpha_2 = -a$ and $\alpha_1\alpha_2 = b$ and β_1 and β_2 be the roots of the equation $x^2 + bx + a = 0$

Therefore, $\beta_1 + \beta_2 = -b$ and $\beta_1\beta_2 = a$

Given that $\alpha_1 - \alpha_2 = \beta_1 - \beta_2$

$$\Rightarrow (\alpha_1 + \alpha_2)^2 - 4\alpha_1\alpha_2 = (\beta_1 + \beta_2)^2 - 4\beta_1\beta_2$$

$$\Rightarrow (-a)^2 - 4b = t - 4a$$

$$\Rightarrow a^2 - b^2 + 4a - 4b = 0$$

$$\Rightarrow (a - b)(a + b) + 4(a - b) = 0$$

$$\Rightarrow (a - b)(a + b + 4) = 0$$

Therefore, $a - b = 0$ and $a + b + 4 = 0$

Question67

For what values of $a \in Z$, the quadratic expression $(x + a)(x + 1991) + 1$ can be factorised as $(x + b)(x + c)$, where $b, c \in Z$?

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Options:

A. 1990

B. 1989

C. 1991

D. 1992

Answer: B

Solution:

$$\text{Given, } (x + a)(x + 1991) + 1$$

$$\therefore (x + a)(x + 1991) - 1 = 0$$

$$\Rightarrow (x + a)(x + 1991) = -1$$

Multiplication of two integers is -1 , this implies that either $(x + a)$ is $+1$ and $(x + 1991)$ is -1 or $(x + a)$ is -1 and $(x + 1991)$ is $+1$.

$$\text{Let } (x + a) = 1, \text{ then } x + 1991 = -1$$

$$\Rightarrow x = -1 - 1991$$

$$x = -1992$$

$$\therefore a = 1 - x = 1 - (-1992) = 1 + 1992 = 1993$$

$$\text{Let } (x + a) = -1, \text{ then } x + 1991 = +1$$

$$\Rightarrow x = 1 - 1991 \\ = -1990$$

$$\therefore a = -1 - x = -1 - (-1990) \\ = -1 + 1990 = 1989$$

So, a can be 1993 or 1989.

Question68

If $\frac{13x+43}{2x^2+17x+30} = \frac{A}{2x+5} + \frac{B}{x+6}$, then $A^2 + B^2 =$

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Options:

A. 22/3

B. 52

C. 34

D. 18/5

Answer: C

Solution:

Given, $\frac{13x+43}{2x^2+17x+30} = \frac{A}{2x+5} + \frac{B}{x+6}$

$$\Rightarrow \frac{13x+43}{2x^2+17x+30} = \frac{A(x+6)+B(2x+5)}{(2x+5)(x+6)}$$

Now, $13x + 43 = (A + 2B)x + (6A + 5B)$

$\therefore A + 2B = 13$ and $6A + 5B = 43$

Solving these equations, we get

$$B = \frac{78-43}{7} = \frac{35}{7} = 5 \text{ and } A = 3$$

$\therefore A^2 + B^2 = (3)^2 + (5)^2 = 9 + 25 = 34$

Question69

If $f(x) = ax^2 + bx + c$ for some $a, b, c \in \mathbb{R}$ with $a + b + c = 3$ and $f(x + y) = f(x) + f(y) + xy, \forall x, y \in \mathbb{R}$. Then, $\sum_{n=1}^{10} f(n) =$

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Options:

A. 330

B. 255

C. 165

D. 190

Answer: A

Solution:

Given: $f(x) = ax^2 + bx + c$ for some $a, b, c \in \mathbb{R}$ with $a + b + c = 3$, and $f(x + y) = f(x) + f(y) + xy$ for all $x, y \in \mathbb{R}$.

Let's analyze the function step by step:

1. Start with the given quadratic function:

$$f(x) = ax^2 + bx + c$$

1. Since $a + b + c = 3$, and given the functional equation $f(x + y) = f(x) + f(y) + xy$, we need to verify and utilize this relationship:

$$f(1) = a + b + c = 3$$

1. Next, consider $f(2)$:

$$f(2) = f(1 + 1) = f(1) + f(1) + 1 = 2f(1) + 1 = 2 \times 3 + 1 = 7$$

1. Now, calculate $f(3)$:

$$f(3) = f(2 + 1) = f(2) + f(1) + 2 = 7 + 3 + 2 = 12$$

1. Then compute $f(4)$:

$$f(4) = f(3 + 1) = f(3) + f(1) + 3 = 12 + 3 + 3 = 18$$

1. For the sum $\sum_{n=1}^{10} f(n)$:

$$\sum_{n=1}^{10} f(n) = f(1) + f(2) + f(3) + \dots + f(10)$$

Let's express each term:

$$f(1) = 3$$

$$f(2) = 2f(1) + 1 = 7$$

$$f(3) = 3f(1) + 3 = 12$$

$$f(4) = 4f(1) + 6 = 18$$

...

$$f(10) = 10f(1) + 45 = 75$$

1. Summarize the formula:

$$\sum_{n=1}^{10} f(n) = f(1) + f(2) + f(3) + \dots + f(10)$$

$$= f(1)[1 + 2 + 3 + \dots + 10] + [1 + 3 + 6 + 10 + 15 + 21 + 28 + 36 + 45]$$

$$= 3 \left(\frac{10(10+1)}{2} \right) + 165$$

$$= 3 \times 55 + 165$$

$$= 165 + 165$$

$$= 330$$

Therefore, the sum $\sum_{n=1}^{10} f(n) = 330$.

Question70

The number of positive real roots of the equation $3^{x+1} + 3^{-x+1} = 10$ is

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Options:

A. 3

B. 2

C. 1

D. Infinitely many

Answer: C

Solution:

$$\begin{aligned}
3^{x+1} + 3^{-x+1} &= 10 \\
\Rightarrow 3^x \cdot 3^1 + 3^{-x} \cdot 3^1 &= 10 \\
\Rightarrow -3y + \frac{3}{y} &= 10 \quad [\text{Let } 3^x = y] \\
\Rightarrow 3y^2 - 10y + 3 &= 0 \\
\Rightarrow 3y^2 - 9y - y + 3 &= 0 \\
\Rightarrow 3y(y - 3) - 1(y - 3) &= 0 \\
\Rightarrow (y - 3)(3y - 1) &= 0 \\
\Rightarrow y = 3 \text{ and } y = \frac{1}{3} \\
\Rightarrow 3^x = 3^1 \text{ and } 3^x = 3^{-1} \\
\Rightarrow x = 1 \text{ and } x = -1
\end{aligned}$$

Number of positive real roots is 1.

Question71

The number of real roots of the equation $\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$ is

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Options:

- A. 1
- B. 2
- C. 3
- D. 4

Answer: B

Solution:

$$\sqrt{\frac{x}{1-x}} + \sqrt{\frac{1-x}{x}} = \frac{13}{6}$$

$$\Rightarrow \frac{x + (1 - x)}{(\sqrt{1 - x})(\sqrt{x})} = \frac{13}{6} \Rightarrow \frac{1}{\sqrt{x - x^2}} = \frac{13}{6}$$

$$\Rightarrow 6 = 13\sqrt{x - x^2}$$

$$\Rightarrow 36 = 169(x - x^2)$$

$$\Rightarrow 169x^2 - 169x + 36 = 0$$

$$D = (-169)^2 - 4(169)(36) = 4225 > 0$$

$$\therefore D > 0$$

$\therefore$ Number of roots will be 2.

Question 72

For $a \neq b$, if the equation $x^2 + ax + b = 0$ and $x^2 + bx + a = 0$ have a common root, then the value of $a + b$ is equal to

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Options:

- A. -1
- B. 0
- C. 1
- D. 2

Answer: A

Solution:

Given, equations are

$$x^2 + ax + b = 0 \quad \dots (i)$$

$$x^2 + bx + a = 0 \quad \dots (ii)$$

Let α be common root of Eqs. (i) and (ii), then

$$\alpha^2 + a\alpha + b = 0 \quad \dots (iii)$$

$$\alpha^2 + b\alpha + a = 0 \quad \dots (iv)$$

Compare Eqs. (iii) and (iv), we get

$$\alpha^2 + a\alpha + b = \alpha^2 + b\alpha + a$$

$$\Rightarrow \alpha(a - b) = (a - b)$$

$$\Rightarrow \alpha = 1$$

Common root = 1

From Eq. (iii), we get

$$1 + a + b = 0$$

$$\Rightarrow a + b = -1$$

Question 73

If the product of the roots of $9x^3 + 112x^2 - 120x + a = 0$ is 12, then the value of a is

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Options:

A. -12

B. 12

C. -108

D. 108

Answer: C

Solution:

Given equation is

$$9x^3 + 112x^2 - 120x + a = 0$$

Let α, β, γ be roots of Eq.(i), then

$$\text{Product of roots} = \alpha\beta\gamma = -\frac{a}{9}$$

$$\text{Given, } -\frac{a}{9} = 12$$

$$\Rightarrow a = -12 \times 9 = -108$$

$$\therefore a = -108$$

Question74

$2 + \sqrt{5}, 1$ are roots of the cubic equation given by
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Options:

A. $x^3 + 3x^2 - 3x - 1 = 0$

B. $x^3 - 3x^2 + 3x - 1 = 0$

C. $x^3 - 5x^2 + 3x + 1 = 0$

D. $x^3 + 5x^2 - 3x + 1 = 0$

Answer: C

Solution:

Roots of cubic equations are 1 and $2 + \sqrt{5}$.

Since, $2 + \sqrt{5}$ is root, then its conjugate also become roots, i.e. $2 - \sqrt{5}$ is root of cubic equation.

$\therefore$ Roots = $1, 2 + \sqrt{5}, 2 - \sqrt{5}$ i.e. α, β, γ

Let cubic equation be

$$x^3 + bx^2 + cx + d = 0$$

Then,

$$\begin{aligned}\text{Sum of Roots} &= -b = 1 + 2 + \sqrt{5} + 2 - \sqrt{5} = 5 \\ \Rightarrow b &= -5\end{aligned}$$

$$\begin{aligned}\text{Product of Roots} &= -d = (1)(2 + \sqrt{5})(2 - \sqrt{5}) \\ &= 4 - 5 = -1 \\ \Rightarrow d &= 1\end{aligned}$$

$$\therefore \alpha\beta + \beta\gamma + \gamma\alpha = c$$

$$(1)(2 + \sqrt{5}) + (2 + \sqrt{5})(2 - \sqrt{5}) + (2 - \sqrt{5})(1) = c$$

$$2 + \sqrt{5} + 2 - \sqrt{5} + 4 - 5 = c$$

$$\Rightarrow$$

$$3 = c$$

$\therefore$ Cubic equation is

$$x^3 - 5x^2 + 3x + 1 = 0$$

Question75

If α and β are the roots of the quadratic equation $x^2 + x + 1 = 0$, then the equation whose roots are $\alpha^{2021}, \beta^{2021}$ is given by

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Options:

A. $x^2 - x + 1 = 0$

B. $x^2 + x - 1 = 0$

C. $x^2 - x - 1 = 0$

D. $x^2 + x + 1 = 0$

Answer: D

Solution:

$$x^2 + x + 1 = 0$$

Roots : $\alpha = \omega$ and $\beta = \omega^2$

where, $\omega = \frac{-1 + \sqrt{3}i}{2}$ and $\omega^3 = 1$

$$\alpha^{2021} = \omega^{2021} = \omega^2$$

and $\beta^{2021} = \omega^{4042} = \omega$

$\therefore$ Equation whose roots are α^{2021} and β^{2021} is

$$(x - \omega)(x - \omega^2) = x^2 + x + 1$$

Question76

If 2, 1 and 1 are roots of the equation $x^3 - 4x^2 + 5x - 2 = 0$, then the roots of $\left(x + \frac{1}{3}\right)^3 - 4\left(x + \frac{1}{3}\right)^2 + 5\left(x + \frac{1}{3}\right) - 2 = 0$

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Options:

A. $\frac{7}{3}, \frac{4}{3}, \frac{4}{3}$

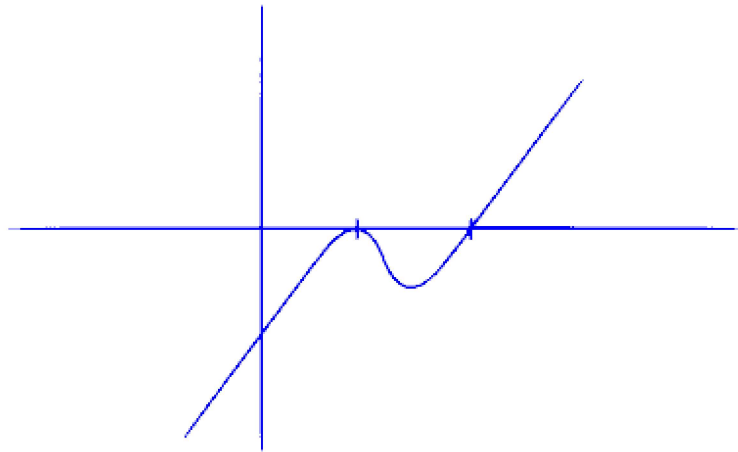
B. $\frac{5}{3}, \frac{2}{3}, \frac{2}{3}$

C. $\frac{-5}{3}, \frac{-2}{3}, \frac{-2}{3}$

D. $\frac{-7}{3}, \frac{-4}{3}, \frac{-4}{3}$

Answer: B

Solution:



This is the graph of $f(x) = x^3 - 4x^2 + 5x - 2$

And the graph of $f\left(x + \frac{1}{3}\right)$ will shift towards left by $\frac{1}{3}$ units, so the new roots will also shift by $\frac{1}{3}$ units towards left.

$$\therefore \text{Roots} = \left(1 - \frac{1}{3}, 1 - \frac{1}{3}, 2 - \frac{1}{3}\right) = \left(\frac{2}{3}, \frac{2}{3}, \frac{5}{3}\right)$$

Question77

If $f(x) = 2x^3 + mx^2 - 13x + n$ and 2, 3 are the roots of the equation $f(x) = 0$, then the values of m and n are

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Options:

A. $-5, -30$

B. $-5, 30$

C. 5, 30

D. 5, -30

Answer: B

Solution:

$$f(x) = 2x^3 + mx^2 - 13x + n$$

Let α, β, γ are the roots of $f(x) = 0$ where $\alpha = 2, \beta = 3$ and let $\gamma = k$

$$\Rightarrow \alpha + \beta + \gamma = -\frac{m}{2} = 2 + 3 + k \quad \dots\dots(i)$$

$$\Rightarrow \alpha\beta + \beta\gamma + \gamma\alpha = -\frac{13}{2} = 6 + 5k \quad \dots\dots(ii)$$

$$\Rightarrow \alpha\beta\gamma = -\frac{n}{2} = 6k \quad \dots\dots(iii)$$

From Eq. (ii), $-13 = 12 + 10k$

$$\Rightarrow k = \frac{-5}{2}$$
$$n = -12k = 30$$
$$m = -10 - 2k = -5$$

Question78

If α and β are the roots of $11x^2 + 12x - 13 = 0$, then $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ is equal to (approximately close to)

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Options:

A. 4.54

B. 3.54

C. 2.54

D. 1.54

Answer: C

Solution:

Given equation, $11x^2 + 12x - 13 = 0$

If α, β are the roots, then

$$\alpha + \beta = \frac{-12}{11} \text{ and } \alpha\beta = -\frac{13}{11}$$

$$\text{Now, } \frac{1}{\alpha^2} + \frac{1}{\beta^2} = \frac{(\alpha+\beta)^2 - 2\alpha\beta}{(\alpha\beta)^2}$$

$$= \frac{\frac{144}{121} + \frac{26}{11}}{\frac{169}{121}}$$

$$= \frac{144 + 286}{169} = 2.54$$

Question79

The value of a for which the equations $x^3 + ax + 1 = 0$ and $x^4 + ax^2 + 1 = 0$ have a common root is

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Options:

A. 2

B. -2

C. 0

D. 1

Answer: B

Solution:

Let α be the common root of

$$x^3 + ax + 1 = 0$$

$$\text{and } x^4 + ax^2 + 1 = 0$$

$$\text{Then, } \alpha^3 + a\alpha + 1 = 0$$

$$\text{and } \alpha^4 + a\alpha^2 + 1 = 0$$

subtracting both equations

$$\begin{aligned}\alpha^4 - \alpha^3 + a\alpha^2 - a\alpha &= 0 \\ \alpha(\alpha^3 - \alpha^2 + a\alpha - a) &= 0 \\ \Rightarrow \alpha^2(\alpha - 1) + a(\alpha - 1) &= 0 \\ [\alpha = 0 \text{ doesn't give a root}] & \\ \Rightarrow (\alpha - 1)(\alpha^2 + a) &= 0 \\ \alpha = 1 \text{ or } a = -\alpha^2 &\end{aligned}$$

Clearly, $\alpha = 1$ gives $a = -2$ in both equations.

Hence, for $a = -2$ both equations have a common root.

Question 80

If a is a positive integer such that roots of the equation $7x^2 - 13x + a = 0$ are rational numbers, then the smallest possible value of a is

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Options:

- A. 5
- B. 6
- C. 7
- D. 8

Answer: B

Solution:

Given, equation, $7x^2 - 13x + a = 0$

$\therefore$ Roots are rational $\Rightarrow$ Discriminate is a perfect square

$$D = 169 - 28a \Rightarrow D = b^2 - 4ac$$

$\Rightarrow D$ is perfect square for $a = 6$

Question81

The sum of the roots of the equation

$$e^{4t} - 10e^{3t} + 29e^{2t} - 22e^t + 4 = 0 \text{ is}$$

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Options:

A. $\log_e 10$

B. $2 \log_e 2$

C. $\log_2 29$

D. $2 \log_{10} 2$

Answer: B

Solution:

$$e^{4t} - 10e^{3t} + 29e^{2t} - 22e^t + 4 = 0 \dots\dots (i)$$

Let $e^t = x$ and roots of (1) be t_1, t_2, t_3, t_4

$$\therefore x^4 - 10x^3 + 29x^2 - 22x + 4 = 0$$

Then, roots are x_1, x_2, x_3, x_4

$$\text{product of roots } x_1 \cdot x_2 \cdot x_3 \cdot x_4 = 4$$

$$\Rightarrow e^{t_1} \cdot e^{t_2} \cdot e^{t_3} \cdot e^{t_4} = 4 \Rightarrow e^{t_1+t_2+t_3+t_4} = 4$$

$$\Rightarrow t_1 + t_2 + t_3 + t_4 = \log_e 4 = \log_e 2^2 = 2 \log_e 2$$

$$\text{Sum of roots} = 2 \log_e 2$$
